Abstract

Neoclassical microeconomic doctrine treats market price as an unobservable, subjective equilibrium scalar ($P = f(S, D)$), asserting that prices reflect voluntary marginal utility and symmetrical competitive market clearing. This paper presents a complete empirical operationalization of the Universal Price Equation (UPE) from Value Physics, constructing forward biophysical prices and reverse-engineering observed commercial Recommended Retail Prices (RRP) across sixty-eight foundational necessities spanning seven essential provisioning sectors in eastern Australia: Food, Agriculture, Poultry, Sugar, Oilseeds, Chemical Inputs, Fertilizers, Liquid Fuels, Power Utilities, Municipal Water/Sanitation, Essential Healthcare, Building Materials, Interior Linings, Rural Reticulation Infrastructure, Spatial Freight Logistics, Agri-Fiber, and Digital Infrastructure. Drawing on official empirical microdata from ABARES, the Australian Bureau of Statistics (ABS), the Australian Energy Market Operator (AEMO), the Australian Competition and Consumer Commission (ACCC), the National Heavy Vehicle Regulator (NHVR), Australian Wool Innovation (AWI), and regional local government authorities, we decompose each price into its physical substrate base (P_b), thermodynamic exergy destruction (P_e), and human metabolic labor difficulty (U, measured in WEST tokens). Solving the inverse UPE for the implied institutional extractive drag fraction (R_a), we establish an objective, non-usurious Recommended Retail Price based on Difficulty (RRP/D). The empirical findings demonstrate that market prices embed an institutional extractive drag (R_a) ranging from 20.8% in owner-operator municipal water cartage up to 79.2% in domestic broadband data transit, establishing that cost-of-living inflation across Australia's core provisioning supply chains is fundamentally an institutional rentier extraction pathology rather than a failure of physical worker productivity.

Keywords: Universal Price Equation , Value Physics , Recommended Retail Price based on Difficulty (RRP/D) , Extractive Drag , Biophysical Economics , Telecommunications Monopoly , Pharmaceutical Pricing , Construction Material Cartels , Agricultural Processing , Infrastructure Pricing , Cement Manufacturing , Agrochemical Cartels , Drywall Duopolies

JEL: D43 , L11 , Q01 , Q40 , R41 , L96 , I11 , Q11 , L74 , L61 , Q12 , L65

Empirical Case Study: Real-World Pricing and RRP/D Fairness Audit — Food, Fuel, Energy, Shelter Materials, Essential Care, Agricultural Inputs, and Digital Infrastructure

Jarrod Lyndsay Hamilton*

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Alethekanon Research Institute, Division 2: Societal Economics & Macro-Micro Simulation, Australia

1 Introduction & Epistemological Critique of Subjective Pricing

Standard neoclassical microeconomic doctrine asserts that market price represents the marginal utility of consumers balanced against the marginal cost of production. In this paradigm, prices serve as costless information vectors guiding optimal resource allocation. However, this framework suffers from three fatal physical and institutional blind spots:

1. **Dimensional Incommensurability:** Neoclassical production functions treat capital, energy, and labor as fungible value aggregates denominated in nominal currency, ignoring the thermodynamic reality that physical goods require irreversible exergy destruction (Georgescu-Roegen, 1971) and finite human metabolic work.

2. **Coercive Asymmetry:** Consumers purchasing essential subsistence goods (such as potable water, staple grain, domestic heating, respiratory medicine, nitrogen fertilizer, herbicides, or shelter materials) face severe biological and legal penalties if they refuse transactions. Market exchanges under existential necessity are coercive rather than Pareto-optimal.

3. **Institutional Rent Extraction:** Corporate oligopolies, financialized intermediaries, and speculative gatekeepers levy substantial tollbooth rents that inflate retail prices far above actual thermodynamic and metabolic reproduction costs (Soddy, 1926; Sraffa, 1960).

Value Physics resolves these pathologies by replacing subjective utility with the **Universal Price Equation (UPE)** (Hamilton, 2026a,b), anchoring economic valuation in physical thermodynamics, ecological regeneration rates, and biological human exertion.

*Corresponding author. Email: jarrod@alethekanon.org

2 Mathematical Architecture: Forward Pricing vs. Reverse RRP Dissection

2.1 The Universal Price Equation (UPE)

The Universal Price Equation models price as an explicit tri-layer biophysical tensor:

$$P = m_1 \cdot m_2 \cdot \gamma(v_{\text{rel}}) \cdot \left[\frac{S \cdot U}{R_n(1 - R_a)} \right] + P_e + P_b \tag{1}$$

where:

- P_e (**Thermodynamic Exergy Base**): Direct thermodynamic energy consumed and destroyed in transformation, heat, electricity, pumping, and haulage (Joules, kWh, or diesel litres converted to local currency equivalents).
- P_b (**Physical Material Base**): Sunk non-energy raw material mass, seed, fertilizer, mineral substrate, chemical reagents, and capital replacement depreciation.
- U (**Metabolic Labor Difficulty**): Human biological work, cognitive attention, and operational skill, quantified in standard WEST Difficulty Tokens (1 WEST $\approx$ baseline metabolic person-hour normalized to local living wage parity).
- S (**Local Scarcity Multiplier**, $S \geq 1.0$): Regional supply constraint, storage buffer deficit factor, or regulatory release buffer.
- R_n (**Environmental Regeneration Capacity**, $0 < R_n \leq 1.0$): Replenishment rate of the ecological or mineral substrate (e.g., pasture regrowth, crop cycles, aquifer recharge, or network duty life).
- R_a (**Institutional Extractive Drag Fraction**, $0 \leq R_a < 1.0$): Unearned speculative rent, financialized interest overhead, and monopolistic margin tolls.
- $\gamma(v_{\text{rel}})$ (**Relativistic Urgency Factor**): Lorentz-dilated coercive pressure factor:

  $$\gamma(v_{\text{rel}}) = \frac{1}{\sqrt{1 - v_{\text{rel}}^2}} \tag{2}$$

  where $v_{\text{rel}} \in [0, 1)$ represents the velocity of economic or biological coercion (e.g., acute respiratory distress, water tank depletion, seasonal weather windows, contractual default penalty).
- m_1, m_2: Wholesale and retail intermediary commercial markup multipliers.

2.2 Reverse Engineering Commercial Retail Prices (RRP)

Given an observed market retail price P_{retail}, we isolate the metabolic spread:

$$\Delta_{\text{spread}} = P_{\text{retail}} - (P_e + P_b) \tag{3}$$

Assuming unaugmented competitive baseline conditions ($m_1 = m_2 = 1.0$), we solve directly for the implied institutional extractive drag fraction:

$$R_a = 1 - \frac{\gamma(v_{\text{rel}}) \cdot S \cdot U}{R_n \cdot \Delta_{\text{spread}}} \tag{4}$$

2.3 Recommended Retail Price based on Difficulty (RRP/D)

The objective, non-usurious baseline price reflecting genuine biophysical cost-of-being is established by evaluating the UPE with a minimal, non-extractive physical contingency reserve ($R_a = 0.02$, representing a 2% emergency buffer stock):

$$\text{RRP/D} = \gamma(v_{\text{rel}}) \cdot \left[\frac{S \cdot U}{R_n(1 - 0.02)} \right] + P_e + P_b \tag{5}$$

2.4 Quantitative Fairness Metrics

We evaluate market performance using two objective metrics:

1. **Extractive Markup Premium ($\Delta P_{\text{extractive}}$):**

$$\Delta P_{\text{extractive}} = \max(0, P_{\text{retail}} - \text{RRP/D}) \tag{6}$$

2. **Biophysical Fairness Index (Φ_{fairness}):**

$$\Phi_{\text{fairness}} = \frac{\text{RRP/D}}{P_{\text{retail}}} \times 100\% \tag{7}$$

A Fairness Index of 100% indicates zero rentier tollbooth extraction. Indices below 80% indicate severe oligopolistic or financialized distortion.

3 Empirical Calibration Methodology & Official Microdata Sources

Empirical parameters were calibrated against verified Australian governmental and industry datasets:

- **Broadacre Grain, Oilseeds, Chemical Inputs & Fertilizer:** ABARES Agricultural Commodities Reports (ABARES, 2024), GrainCorp port schedules, Incitec Pivot / CSBP pricing, Nufarm / Corteva product schedules, and NSW DPI Farm Gross Margin Budgets.
- **Retail Food & Grocery Margins:** ACCC Supermarkets Inquiry Reports (ACCC, 2024) and ABS Consumer Price Index (ABS, 2024a).
- **Dairy, Poultry & Sugar Processing:** Dairy Australia Situation & Outlook (Dairy Australia, 2024), Australian Chicken Meat Federation benchmarks, and Canegrowers/QSL annual reports.
- **Livestock & Pastoral Production:** Meat & Livestock Australia (MLA) Eastern Young Cattle Indicator and NSW DPI Beef Gross Margins.
- **Agri-Fiber & Wool Markets:** Australian Wool Innovation (AWI) production benchmarks and AWEX Eastern Market Indicator microdata.
- **Liquid Fuels & Refining:** Australian Institute of Petroleum (AIP) Terminal Gate Pricing and ACCC Petroleum Monitoring Reports (ACCC, 2025).

- **Electricity Generation & Transmission:** AEMO Quarterly Energy Dynamics (AEMO, 2025) and AER Default Market Offer.
- **Municipal Water & Waste:** Regional Local Government Council Operational Plans (Warrumbungle Shire Council, 2025), NSW Water Directorate benchmarking, and NSW EPA Waste Levy Guidelines.
- **Pharmaceutical Healthcare:** Department of Health Pharmaceutical Benefits Scheme (PBS), Therapeutic Goods Administration (TGA) manufacturing audits, and Pharmacy Guild dispensing schedules.
- **Building Materials, Linings, Steel, Cement & Infrastructure:** ABS Producer Price Indexes (ABS, 2024b), Master Builders Australia, Cement Industry Federation (CIF), CSR Gyprock, Knauf, BlueScope/Lysaght, and Iplex/Vinidex catalogs.
- **Road Freight Logistics:** National Heavy Vehicle Regulator (NHVR) regulations (NHVR, 2024) and Newell Highway commercial rate cards.
- **Digital Telecommunications:** ACCC Wholesale Telecommunications Market Reports and NBN Co Special Access Undertaking (SAU) pricing schedules.

4 Fifty-Six Granular Empirical Case Studies

4.1 Sector 1: Food, Agriculture, Pastoral Livestock, Poultry & Crops

Case 1: Commercial Sliced White Bread (680g Loaf) Observed RRP: \$4.10 AUD. Inputs: $P_e = \$0.38$, $P_b = \$0.45$, $U = 2.15$, $S = 1.00$, $R_n = 1.00$, $v_{\text{rel}} = 0.05$. Fair RRP/D = **\$3.03 AUD**. Implied extractive drag $R_a = \mathbf{34.2}\%$, imposing an extractive premium of \$1.07/loaf (Fairness Index: **73.8%**). Supermarket duopoly shelf slotting fees and retail property yields capture the dominant share of consumer expenditure.

Case 2: Milling Wheat (APW1 / ASW1 per Tonne) Observed Spot Price: \$365.00 AUD/t. Inputs: $P_e = \$104.40$, $P_b = \$82.00$, $U = 95.00$, $S = 1.05$, $R_n = 0.98$, $v_{\text{rel}} = 0.02$. Fair RRP/D = **\$290.28 AUD/t**. Implied extractive drag $R_a = \mathbf{43.0}\%$, with an extractive premium of \$74.72/t (Fairness Index: **79.5%**) captured by grain bulk handling and export port cartels.

Case 3: Fresh Full Cream Milk (Full Fat, 1 Litre) Observed RRP: \$2.10 AUD/L. Inputs: $P_e = \$0.31$, $P_b = \$0.48$, $U = 0.75$, $S = 1.05$, $R_n = 0.95$, $v_{\text{rel}} = 0.10$. Fair RRP/D = **\$1.64 AUD/L**. Implied extractive drag $R_a = \mathbf{36.4}\%$, with an extractive premium of \$0.46/L (Fairness Index: **78.1%**). Supermarket private-label discounting and processor duopsonies compress farmgate returns to $\approx$ \$0.68/L while extracting unearned retail margins.

Case 4: Farmgate Beef Cattle Weaners (British Breed Steers, per kg Liveweight) Observed Saleyard Price: \$3.85 AUD/kg lwt (\$1,155/300kg steer). Inputs: $P_e = \$0.42$, $P_b = \$1.15$, $U = 1.25$, $S = 1.15$, $R_n = 0.85$, $v_{\text{rel}} = 0.15$. Fair RRP/D = **\$3.32 AUD/kg lwt**. Implied extractive drag $R_a = \mathbf{25.0}\%$, with an extractive premium of \$0.53/kg (Fairness Index: **86.1%**). Producer livestock auctions reflect high physical labor and pasture exergy fidelity.

Case 5: Australian Merino Greasy Wool (19-Micron Fleece, per kg Clean) Observed Auction Benchmark: \$15.50 AUD/kg clean. Inputs: $P_e = \$1.45$, $P_b = \$2.20$, $U = 5.60$, $S = 1.10$, $R_n = 0.88$, $v_{\text{rel}} = 0.08$. Fair RRP/D = **\$10.82 AUD/kg**. Implied extractive drag $R_a = \mathbf{40.7}\%$, imposing an extractive premium of \$4.68/kg (Fairness Index: **69.8%**). While physical shearing exertion, crutching, and pastoral exergy form the core difficulty floor, auction house brokerage commissions, core testing charges, and export financing capture $\approx 40.7\%$ in unearned institutional intermediation drag.

Case 6: Commercial Broiler Chicken (Whole Fresh Dressed Bird, per kg) Observed Supermarket RRP: \$4.50 AUD/kg dressed. Inputs: $P_e = \$0.46$, $P_b = \$1.68$, $U = 1.15$, $S = 1.05$, $R_n = 0.95$, $v_{\text{rel}} = 0.12$. Fair RRP/D = **\$3.45 AUD/kg**. Implied extractive drag $R_a = \mathbf{45.8}\%$, imposing an extractive premium of \$1.05/kg (Fairness Index: **76.6%**). Feed milling and shed ventilation exergy represent \$0.46/kg, and grain feed/chick input costs represent \$1.68/kg. The 45.8% extractive drag reflects the concentrated processor duopsony (Inghams/Baiada) and retail supermarket margin stacking.

Case 7: Coarse Solar Sea Salt (Unrefined Food & Livestock Mineral Feedstock, 25kg Bag) Observed Rural Merchant Price: \$14.50 AUD/25kg bag. Inputs: $P_e = \$1.65$, $P_b = \$2.10$, $U = 4.20$, $S = 1.05$, $R_n = 0.96$, $v_{\text{rel}} = 0.06$. Fair RRP/D = **\$8.45 AUD/25kg bag**. Implied extractive drag $R_a = \mathbf{57.2}\%$, imposing an extractive premium of \$6.05/bag (Fairness Index: **58.2%**). Sourced from Cheetham Salt and Olsson's production data, natural coastal solar evaporation ponds require minimal thermal exergy ($P_e + P_b = \$3.75$), yet commercial distributor packaging and wholesale merchant tollbooths capture an implied 57.2% extractive drag.

Case 8: Raw Centrifugal Cane Sugar (Bulk Milling Crystal Feedstock, per Tonne) Observed Wholesale Benchmark: \$620.00 AUD/t. Inputs: $P_e = \$142.00$, $P_b = \$168.00$, $U = 145.00$, $S = 1.08$, $R_n = 0.90$, $v_{\text{rel}} = 0.05$. Fair RRP/D = **\$487.77 AUD/t**. Implied extractive drag $R_a = \mathbf{43.8}\%$, imposing an extractive premium of \$132.23/t (Fairness Index: **78.7%**). Grounded in Canegrowers and QSL pool data, sugarcane harvesting and mill boiler exergy represent \$142.00/t, while mill consolidation and port terminal access charges capture an implied 43.8% extractive toll.

Case 9: Farmgate Canola Grain (Bulk Non-GM Oilseed, per Tonne) Observed Silo Benchmark: \$685.00 AUD/t. Inputs: $P_e = \$118.50$, $P_b = \$215.00$, $U = 112.00$, $S = 1.08$, $R_n = 0.92$, $v_{\text{rel}} = 0.04$. Fair RRP/D = **\$467.77 AUD/t**. Implied extractive drag $R_a = \mathbf{62.6}\%$, imposing an extractive premium of \$217.23/t (Fairness Index: **68.3%**). Sourced from GrainCorp, AGFlow, and ABARES datasets, high input costs (synthetic nitrogen, hybrid seeds, fungicides) require $P_e + P_b = \$333.50/t$. International oilseed trading cartels and export terminal access markups extract an implied 62.6% drag, leaving growers exposed to extreme input volatility.

Case 10: Bulk Granular Agricultural Urea Fertilizer (46% Nitrogen, per Tonne) Observed Depot Price: \$740.00 AUD/t. Inputs: $P_e = \$315.00$, $P_b = \$165.00$, $U = 78.00$, $S = 1.14$, $R_n = 0.84$, $v_{\text{rel}} = 0.15$. Fair RRP/D = **\$589.25 AUD/t**. Implied extractive drag $R_a = \mathbf{58.8}\%$, imposing an extractive premium of \$150.75/t (Fairness Index: **79.6%**). Grounded in Incitec Pivot and World Bank benchmarks, Haber-Bosch steam methane reforming natural gas exergy and fluid-bed granulation require $P_e + P_b = \$480.00/t$. Fertilizer import cartels and wholesale distributor markups extract an implied 58.8% toll, inflating input costs across Australian agriculture.

Case 11: Agricultural Glyphosate Herbicide (450g/L SL Systemic Broadacre Formulation, per 20L Drum) Observed Rural Merchant Price: \$135.00 AUD/drum. Inputs: $P_e = \$22.50$, $P_b = \$54.00$, $U = 21.50$, $S = 1.12$, $R_n = 0.86$, $v_{\text{rel}} = 0.18$. Fair RRP/D = **\$105.55 AUD/drum**. Implied extractive drag $R_a = \mathbf{51.3}\%$, imposing an extractive premium of \$29.45/drum (Fairness Index: **78.2%**). Sourced from Nufarm and APVMA regulatory benchmarks, technical active synthesis exergy and HDPE drum blow-molding require $P_e + P_b = \$76.50/drum$. Agrochemical multinational distribution cartels extract an implied 51.3% drag, compounding input costs on direct-drill farming.

4.2 Sector 2: Energy & Power Utilities

Case 12: Automotive Diesel Fuel (ULSD per Litre) Observed RRP: \$2.15 AUD/L. Inputs: $P_e = \$0.32$, $P_b = \$0.92$, $U = 0.35$, $S = 1.08$, $R_n = 0.90$, $v_{\text{rel}} = 0.12$. Fair RRP/D = **\$1.67 AUD/L**. Implied extractive drag $R_a = \mathbf{53.5}\%$, imposing an extractive premium of \$0.48/L (Fairness Index: **77.8%**).

Case 13: Residential Grid Electricity (Single-Rate Flat per kWh) Observed Tariff: \$0.36 AUD/kWh. Inputs: $P_e = \$0.08$, $P_b = \$0.07$, $U = 0.09$, $S = 1.15$, $R_n = 0.92$, $v_{\text{rel}} = 0.08$. Fair RRP/D = **\$0.27 AUD/kWh**. Implied extractive drag $R_a = \mathbf{46.3}\%$, with an extractive premium of \$0.09/kWh (Fairness Index: **73.7%**).

Case 14: LPG Bottled Gas (45kg Domestic Cylinder) Observed Delivered RRP: \$165.00 AUD/cylinder. Inputs: $P_e = \$29.40$, $P_b = \$58.36$, $U = 28.50$, $S = 1.15$, $R_n = 0.88$, $v_{\text{rel}} = 0.16$. Fair RRP/D = **\$126.26 AUD**. Implied extractive drag $R_a = \mathbf{51.1}\%$, with an extractive premium of \$38.74/cylinder (Fairness Index: **76.5%**) extracted by regional distribution duopolies.

4.3 Sector 3: Municipal Water, Sanitation & Essential Healthcare

Case 15: Bulk Potable Water Cartage (Regional NSW, per 1,000 Litres / 1 kL) Observed Delivery Price: \$15.50 AUD/kL (\$155.00 per 10kL load). Inputs: $P_e = \$4.32$, $P_b = \$3.95$, $U = 3.20$, $S = 1.30$, $R_n = 0.75$, $v_{\text{rel}} = 0.25$. Fair RRP/D = **\$14.12 AUD/kL**. Implied extractive drag $R_a = \mathbf{20.8}\%$, yielding the highest Fairness Index in the study (**91.1%**). Because water cartage is conducted by independent owner-drivers facing high physical road diesel exergy and driving exertion, retail prices track genuine biophysical cost-of-being.

Case 16: Urban Reticulated Potable Water (Municipal Pipeline Tariff, Regional NSW per kL) Observed Volumetric Tariff: \$3.60 AUD/kL. Inputs: $P_e = \$0.48$, $P_b = \$0.62$, $U = 1.15$, $S = 1.15$, $R_n = 0.80$, $v_{\text{rel}} = 0.20$. Fair RRP/D = **\$2.82 AUD/kL**. Implied extractive drag $R_a = \mathbf{32.5}\%$, imposing an extractive premium of \$0.78/kL (Fairness Index: **78.4%**). Water treatment filtration electricity and river pumping exergy require $P_e = \$0.48$, while coagulant chemicals and mains renewal depreciation account for $P_b = \$0.62$. Council administrative overhead and dividend extraction account for the remaining 32.5% spread.

Case 17: Domestic Waste Collection (Regional NSW, per Weekly Bin Lift) Observed Council Charge: \$9.33 AUD/lift (\$485.00 annual charge / 52 weeks). Inputs: $P_e = \$2.45$, $P_b = \$1.85$, $U = 2.15$, $S = 1.20$, $R_n = 0.72$, $v_{\text{rel}} = 0.22$. Fair RRP/D = **\$8.05 AUD/lift**. Implied extractive drag $R_a = \mathbf{27.0}\%$ (Fairness Index: **86.3%**). Non-profit municipal provision prevents monopolistic dividend extraction.

Case 18: Prescription Antibiotic (Amoxicillin 500mg, 20 Capsules Course) Observed Private Retail Price: \$16.50 AUD/course. Inputs: $P_e = \$2.10$, $P_b = \$3.20$, $U = 5.80$, $S = 1.20$, $R_n = 0.94$, $v_{\text{rel}} = 0.35$. Fair RRP/D = **\$13.37 AUD/course**. Implied extractive drag $R_a = \mathbf{29.4}\%$, with an extractive premium of \$3.13/course (Fairness Index: **81.0%**). Professional pharmacist clinical verification and sterile fermentation represent the legitimate core of value.

Case 19: Prescription Salbutamol Inhaler (100mcg CFC-Free MDI, 200 Actuations) Observed Private Retail Price: \$12.50 AUD/inhaler. Inputs: $P_e = \$0.65$, $P_b = \$1.35$, $U = 3.40$, $S = 1.10$, $R_n = 0.92$, $v_{\text{rel}} = 0.40$. Fair RRP/D = **\$6.53 AUD/inhaler**. Implied extractive drag $R_a = \mathbf{57.8}\%$, imposing an extractive premium of \$5.97/inhaler (Fairness Index: **52.2%**). Grounded in TGA good manufacturing practice (GMP) audits and PBS item schedules, active micronized salbutamol sulfate powder and medical HFA propellant represent modest physical base costs ($P_e + P_b = \$2.00$). However, because acute bronchospasm creates extreme biological urgency ($v_{\text{rel}} = 0.40, \gamma = 1.0911$), multinational brand patents and pharmacy retail distribution extract an implied 57.8% extractive drag on life-critical respiratory management.

4.4 Sector 4: Shelter, Building Materials, Rural Infrastructure, Packaging, Logistics & Telecom

Case 20: Residential Softwood Framing Timber (MGP10 90x45mm Pine per Metre) Observed RRP: \$7.20 AUD/m. Inputs: $P_e = \$0.85$, $P_b = \$1.20$, $U = 2.10$, $S = 1.25$, $R_n = 0.88$, $v_{\text{rel}} = 0.15$. Fair RRP/D = **\$5.13 AUD/m**. Implied extractive drag $R_a = \mathbf{41.4}\%$, with an extractive premium of \$2.07/m (Fairness Index: **71.2%**).

Case 21: Pre-Mixed Ready-Mix Concrete (N20 20 MPa Structural per m^3) Observed RRP: \$295.00 AUD/m^3. Inputs: $P_e = \$62.00$, $P_b = \$78.00$, $U = 58.00$, $S = 1.15$, $R_n = 0.85$, $v_{\text{rel}} = 0.20$. Fair RRP/D = **\$221.72 AUD/m^3**. Implied extractive drag $R_a = \mathbf{48.3}\%$, with an extractive premium of \$73.28/m^3 (Fairness Index: **75.2%**).

Case 22: Structural Steel Reinforcing Mesh (SL82, 6.0m x 2.4m Sheet) Observed Trade RRP: \$128.00 AUD/sheet. Inputs: $P_e = \$28.50$, $P_b = \$36.80$, $U = 22.50$, $S = 1.12$, $R_n = 0.82$, $v_{\text{rel}} = 0.18$. Fair RRP/D = **\$97.18 AUD/sheet**. Implied extractive drag $R_a = \mathbf{50.2}\%$, with an extractive premium of \$30.82/sheet (Fairness Index: **75.9%**).

Case 23: Standard Extruded Clay Facing Bricks (Common Structural Masonry per 1,000 Units) Observed Trade Price: \$1,150.00 AUD/1,000 bricks. Inputs: $P_e = \$278.00$, $P_b = \$182.00$, $U = 185.00$, $S = 1.12$, $R_n = 0.82$, $v_{\text{rel}} = 0.18$. Fair RRP/D = **\$722.12 AUD/1,000 bricks**. Implied extractive drag $R_a = \mathbf{62.8}\%$, with an extractive premium of \$427.88/1,000 bricks (Fairness Index: **62.8%**). High-temperature natural gas tunnel kiln firing (1,050°C–1,150°C) consumes \$278.00 in thermal and machinery exergy, while clay quarry extraction and refractory wear require \$182.00. The commercial brick manufacturing duopoly (Brickworks/CSR) extracts a substantial 62.8% drag on structural walling materials.

Case 24: Standard Gypsum Plasterboard Wall Lining (10mm Recessed Edge, 1.2m x 2.4m Sheet) Observed Trade Price: \$24.80 AUD/sheet. Inputs: $P_e = \$4.90$, $P_b = \$6.40$, $U = 4.10$, $S = 1.10$, $R_n = 0.85$, $v_{\text{rel}} = 0.14$. Fair RRP/D = **\$16.77 AUD/sheet**. Implied extractive drag $R_a = \mathbf{60.3}\%$, imposing an extractive premium of \$8.03/sheet (Fairness Index: **67.6%**). Grounded in CSR Gyprock and Knauf benchmarks, gypsum calcination and board drying kilns require $P_e + P_b = \$11.30/sheet$. The domestic drywall duopoly extracts an implied 60.3% drag, confirming that interior linings mirror the severe extraction of structural envelope materials.

Case 25: Commercial Corrugated Galvanized Steel Roofing Sheets (Custom Orb 0.42mm BMT) Observed Trade Price: \$21.50 AUD/m. Inputs: $P_e = \$4.85$, $P_b = \$6.20$, $U = 3.80$, $S = 1.10$, $R_n = 0.82$, $v_{\text{rel}} = 0.18$. Fair RRP/D = **\$16.34 AUD/m**. Implied extractive drag $R_a = \mathbf{50.4}\%$, imposing an extractive premium of \$5.16/m (Fairness Index: **76.0%**). Blast furnace smelting, galvanizing zinc coating (Z275), and roll-forming exergy represent $P_e + P_b = \$11.05/m$. BlueScope flat steel monopoly pricing and distributor network margins extract an implied 50.4% toll, completing the residential structural building envelope audit.

Case 26: Commercial HDPE Rural Water Reticulation Pipe (32mm PN12.5, 100m Coil) Observed Trade Price: \$245.00 AUD/coil. Inputs: $P_e = \$48.50$, $P_b = \$68.00$, $U = 38.00$, $S = 1.10$, $R_n = 0.88$, $v_{\text{rel}} = 0.16$. Fair RRP/D = **\$165.60 AUD/coil**. Implied extractive drag $R_a = \mathbf{62.5}\%$, imposing an extractive premium of \$79.40/coil (Fairness Index: **67.6%**). Grounded in Vinidex and Iplex price schedules, polymer extrusion electric motors and virgin PE100 resin pellets represent $P_e + P_b = \$116.50/coil$. Concentrated plastic pipe duopolies extract an implied 62.5% drag on essential rural livestock water conveyance.

Case 27: Commercial FIBC Polypropylene Bulk Bags (1-Tonne Woven Container) Observed Merchant Price: \$24.50 AUD/bag. Inputs: $P_e = \$4.20$, $P_b = \$5.90$, $U = 4.80$, $S = 1.08$, $R_n = 0.90$, $v_{\text{rel}} = 0.12$. Fair RRP/D = **\$16.02 AUD/bag**. Implied extractive drag $R_a = \mathbf{59.7}\%$, imposing an extractive premium of \$8.48/bag (Fairness

Index: **65.4%**). Grounded in rural packaging trade schedules, woven polymer tape extrusion and industrial sewing require $P_e + P_b = \$10.10/bag$. Rural merchant and packaging distributor networks extract an implied 59.7% drag on agricultural bulk handling.

Case 28: Bulk Portland Cement (General Purpose GP Clinker, per Tonne) Observed Trade / Silo Gate Price: \$345.00 AUD/tonne. Inputs: $P_e = \$112.00$, $P_b = \$84.00$, $U = 56.00$, $S = 1.12$, $R_n = 0.78$, $v_{\text{rel}} = 0.15$. Fair RRP/D = **\$278.99 AUD/tonne**. Implied extractive drag $R_a = \mathbf{45.4}\%$, imposing an extractive premium of \$66.01/tonne (Fairness Index: **80.9%**). High-temperature rotary kiln clinkerization (1,450°C) consumes \$112.00 in thermal and grinding exergy, while limestone extraction and gypsum require \$84.00. Cement manufacturer oligopolies (Adbri, Boral, Cement Australia) levy an implied 45.4% drag that cascades directly downstream into ready-mix structural concrete ($R_a = 48.3\%$) and civil construction.

Case 29: Bulk Road Freight Logistics (Newell Highway B-Double Corridor) Observed Freight Rate: \$0.1250 AUD/nt-km (\$5.00/km loaded). Inputs: $P_e = \$0.0345$, $P_b = \$0.0285$, $U = 0.0210$, $S = 1.10$, $R_n = 0.92$, $v_{\text{rel}} = 0.14$. Fair RRP/D = **\$0.0889 AUD/nt-km**. Implied extractive drag $R_a = \mathbf{59.1}\%$, with an extractive premium of \$0.0361/nt-km (Fairness Index: **71.1%**). Corporate 3PL broker platforms capture over half of total freight billing.

Case 30: Domestic Broadband Data Transit (NBN 50/20 Mbps Tier, Monthly Service) Observed Retail Plan: \$85.00 AUD/month. Inputs: $P_e = \$4.80$, $P_b = \$14.50$, $U = 12.20$, $S = 1.05$, $R_n = 0.95$, $v_{\text{rel}} = 0.15$. Fair RRP/D = **\$33.22 AUD/month**. Implied extractive drag $R_a = \mathbf{79.2}\%$, imposing an extractive premium of \$51.78/month (Fairness Index: **39.1%**). Grounded in ACCC Wholesale Telecommunications Market Reports and NBN Special Access Undertaking (SAU) filings, photonic transit electricity and routing exergy constitute $P_e = \$4.80$, while optical line terminal hardware depreciation accounts for $P_b = \$14.50$. The unprecedented 79.2% extractive drag fraction establishes that wholesale broadband monopoly pricing and retail telecommunications duopoly packaging extract severe unearned financialized rents on non-discretionary civic connectivity.

Case 31: Commercial Structural Hardwood Timber (F17 Kiln-Dried Hardwood, 90x45mm) Observed Merchant Price: \$14.80 AUD/lineal metre. Inputs: $P_e = \$2.15$, $P_b = \$2.85$, $U = 3.20$, $S = 1.25$, $R_n = 0.72$, $v_{\text{rel}} = 0.14$. Fair RRP/D = **\$10.73 AUD/m**. Implied extractive drag $R_a = \mathbf{42.8}\%$, imposing an extractive premium of \$4.07/m (Fairness Index: **72.5%**). Grounded in Bowens, Melbourne Timber Supplies, and ABS Producer Price Indexes (Cat. 6427.0). Structural hardwood sawmilling requires heavy felling and snigging diesel exergy ($P_e = \$2.15$), extended multi-week kiln dehumidification, and sustainable forestry royalties ($P_b = \$2.85$). Due to native timber harvesting policy transitions across eastern Australia, high scarcity ($S = 1.25$) and long eucalypt biological renewal cycles ($R_n = 0.72$) elevate biophysical difficulty. However, merchant trade distribution networks capture an implied 42.8% extractive drag, directly echoing plantation softwood framing ($R_a = 41.4\%$) and proving that structural framing lumber inflation across both hardwood and softwood species is dominated by wholesale distributor margins.

Case 32: Agricultural Feed Barley (F1 Broadacre Feed Grain, per Tonne) Observed Commercial Spot Price: \$325.00 AUD/tonne. Inputs: P_e = \$92.50, P_b = \$74.00, U = 84.00, S = 1.06, R_n = 0.98, v_{rel} = 0.04. Fair RRP/D = **\$259.29 AUD/tonne**. Implied extractive drag R_a = **42.6**%, establishing an extractive premium of \$65.71/tonne (Fairness Index: **79.8%**). Grounded in ABARES Weekly Commodity Price Updates and GrainCentral regional market indices. Feed barley is the foundational energetic carbohydrate feedstock for domestic beef cattle feedlots, intensive dairy rations, and regional drought supplementary feeding. While tractor tillage, planting, and harvesting exergy (P_e = \$92.50) and seed/basal fertilizer inputs (P_b = \$74.00) establish a biophysical reproduction baseline of \$259.29/t, commercial grain accumulation cartels and bulk handling terminal networks capture an implied 42.6% extractive drag. This closely matches milling wheat (R_a = 43.0%, Fairness 79.5%), demonstrating that broadacre cereal grains face an invariant 43% institutional intermediation toll regardless of end-use processing.

Case 33: Commercial Copper Electrical Building Wire (2.5mm² Twin & Earth Flat TPS Cable) Observed Trade/Retail Price: \$225.00 AUD per 100m drum. Inputs: P_e = \$24.50, P_b = \$118.50, U = 22.00, S = 1.15, R_n = 0.82, v_{rel} = 0.18. Fair RRP/D = **\$175.01 AUD/drum**. Implied extractive drag R_a = **61.8**%, imposing an extractive premium of \$49.99/drum (Fairness Index: **77.8%**). Grounded in Rexel, Middy's Electrical, and Bunnings trade pricing. Australian standard 2.5mm² 2C+Earth flat cable embeds 6.72 kg of electrolytic tough pitch (ETP) refined copper cathode (P_b copper value $\approx$ \$98.80) and 5.5 kg of electrical grade flame-retardant V-90 PVC compound. Tandem diamond die wire-drawing, continuous resistance annealing at 400°C, and dual-layer PVC cross-head extrusion consume P_e = \$24.50. Driven by cable manufacturing oligopolies (Prysmian/Olex) and concentrated electrical merchant distribution, commercial trade pricing embeds an implied 61.8% extractive drag (\$49.99/drum premium), demonstrating that essential electrical infrastructure carries markups comparable to structural plumbing pipe (R_a = 62.5%).

Case 34: Solvent-Extracted Canola Meal / Protein Cake (Bulk Pelletized, 36%–38% Crude Protein) Observed Commercial Feedlot Price: \$580.00 AUD/tonne. Inputs: P_e = \$84.50, P_b = \$212.00, U = 82.00, S = 1.10, R_n = 0.94, v_{rel} = 0.10. Fair RRP/D = **\$394.91 AUD/tonne**. Implied extractive drag R_a = **66.0**%, establishing an extractive premium of \$185.09/tonne (Fairness Index: **68.1%**). Grounded in GrainCentral, Beef Central Feedgrain Focus, and Australian Oilseeds Federation (AOF) benchmarks. Canola meal is the foundational protein feed supplement for eastern Australian beef cattle feedlots and intensive dairy rations. Mechanical expeller pressing, countercurrent hexane solvent extraction, desolventizer-toaster stripping steam (P_e = \$84.50), and allocated raw seed mass (P_b = \$212.00) yield an objective biophysical baseline of \$394.91/t. Global agribusiness crushing houses (GrainCorp, Cargill, Riverina Oils) extract an implied 66.0% drag (\$185.09/t premium), closely corroborating the extractive drag observed in farmgate canola grain (R_a = 62.6%, Fairness 68.3%) and proving that processing oligopolies levy severe rents on essential intensive livestock nutrition.

Case 35: Commercial Hot-Rolled Structural Steel Universal Beams (150UB14.0 Grade 300PLUS) Observed Commercial Trade Price: \$52.00 AUD/lineal metre. Inputs: P_e = \$11.20, P_b = \$14.50, U = 8.20, S = 1.12, R_n = 0.82, v_{rel} = 0.16. Fair

RRP/D = **$37.28 AUD/m**. Implied extractive drag R_a = **56.9**%, imposing an extractive premium of $14.72/m (Fairness Index: **71.7%**). Grounded in InfraBuild, BlueScope Distribution, and Melsteel trade price schedules. Heavy structural steel universal beams (14.0 kg/m) are the essential load-bearing lintels and structural members in multi-storey and wide-span residential construction. Continuous billet casting, reheating furnace gas combustion at 1,250°C, and heavy universal roughing and edging mill rolling consume P_e = $11.20. Raw structural steel scrap, Si-Mn micro-alloying for Grade 300PLUS yield strength, roll wear amortization, and shop primer coating constitute P_b = $14.50. Governed by Australia's concentrated domestic structural steel supply duopoly and merchant distributor price-setting, commercial trade prices embed an implied 56.9% extractive drag ($14.72/m premium), showing that heavy structural shapes carry unearned margins consistent with reinforcing mesh (R_a = 50.2%) and steel roofing (R_a = 50.4%).

Case 36: Whole Ginned Cottonseed (Fuzzy White Cattle Feedstock, per Tonne) Observed Commercial Spot Price: $520.00 AUD/tonne. Inputs: P_e = $86.50, P_b = $178.50, U = 68.00, S = 1.14, R_n = 0.90, v_{rel} = 0.10. Fair RRP/D = **$353.33 AUD/tonne**. Implied extractive drag R_a = **66.1**%, establishing an extractive premium of $166.67/tonne (Fairness Index: **67.9%**). Grounded in Beef Central Feedgrain Focus, GrainCentral, and Namoi Cotton regional gin-gate price benchmarks across the Gwydir, Macquarie, and Darling Downs valleys. Whole fuzzy white cottonseed is the premier whole-oilseed energy and bypass-protein supplement for beef cattle feedlots and pastoral drought feeding (22% crude protein, 18% oil). Tractor harvesting exergy, saw-ginning mechanical power, gin drying tower gas heat, and bulk tipper freight consume P_e = $86.50, while allocated seed crop production costs (after lint revenue allocation) account for P_b = $178.50. Agribusiness grain trading cartels and ginning aggregators extract an implied 66.1% drag ($166.67/t premium), perfectly matching Solvent-Extracted Canola Meal (R_a = 66.0%, Fairness 68.1%) and proving that non-discretionary intensive livestock protein feeds face an invariant 66% rentier extraction toll across eastern Australia.

Case 37: Commercial Cold-Formed Steel Galvanized Framing / C-Purlin (C15015, 1.5mm BMT) Observed Commercial Trade Price: $16.50 AUD/lineal metre. Inputs: P_e = $3.40, P_b = $5.10, U = 2.60, S = 1.10, R_n = 0.82, v_{rel} = 0.16. Fair RRP/D = **$12.11 AUD/m**. Implied extractive drag R_a = **55.8**%, imposing an extractive premium of $4.39/m (Fairness Index: **73.4%**). Grounded in Stramit, Lysaght, Stratco, and regional discount roofing suppliers. Roll-formed from high-tensile G450 galvanized steel strip (2.89 kg/m), continuous zinc pot galvanizing thermal exergy (450°C), cold-roll coil slitting, multi-stand roll-forming drives, and automated hydraulic hole punching consume P_e = $3.40. Strip coil feedstock and Z350/Z275 zinc coating constitute P_b = $5.10. Concentrated roll-forming manufacturer networks and merchant distributors extract an implied 55.8% extractive drag ($4.39/m premium), aligning with corrugated roofing (R_a = 50.4%) and structural universal beams (R_a = 56.9%) to confirm that roll-formed steel framing components embed substantial markups over physical manufacturing difficulty.

Case 38: Australian Prime Hard Breadmaking Wheat (APH2 High-Protein Milling Grain, min 13% Protein) Observed Commercial Depot Price: $410.00 AUD/tonne. Inputs: P_e = $118.50, P_b = $96.50, U = 108.00, S = 1.08, R_n = 0.98,

$v_{\text{rel}} = 0.04$. Fair RRP/D = **$336.55 AUD/tonne**. Implied extractive drag $R_a =$ **38.9**%, establishing an extractive premium of $73.45/tonne (Fairness Index: **82.1%**). Grounded in GrainCentral, ABARES, and GrainCorp regional sub-terminal cash bids across the inner and outer slopes of NSW. Prime Hard wheat requires intensive agronomic management and heavy nitrogen fertilization (120–150 kg N/ha) to achieve min 13.0% grain protein without falling number weather damage, consuming $P_e = \$118.50$ in tractor fuel and in-crop spraying and $P_b = \$96.50$ in fertilizer and seed inputs. While the market price ($410.00/t) exceeds standard milling wheat ($365.00/t), the higher physical base and labor exertion ($U = 108.00$ WEST) yield an extractive drag of 38.9% and an elevated Fairness Index of 82.1%, demonstrating that genuine physical difficulty and nitrogen exergy absorb most of the premium over standard APW1 wheat.

Case 39: Commercial Aluminum Window Glazing Frames (Double-Glazed Residential Sliding Window, 1200mm × 1800mm Standard Unit) Observed Commercial Trade/Retail Price: $540.00 AUD/unit (2.16 m^2). Inputs: $P_e = \$52.00$, $P_b = \$168.00$, $U = 64.80$, $S = 1.14$, $R_n = 0.84$, $v_{\text{rel}} = 0.18$. Fair RRP/D = **$311.23 AUD/unit**. Implied extractive drag $R_a =$ **72.1**%, imposing an extractive premium of $228.77/unit (Fairness Index: **57.6%**). Grounded in architectural fenestration trade schedules from major Australian fabricators (Bradnam's, Stegbar, G.James) for a standard residential sliding window (1200 mm × 1800 mm) comprising 6063-T5 powder-coated extruded aluminum frames (11.5 kg Al) and an insulated glass unit (IGU: 4 mm clear float / 12 mm argon / 4 mm clear float, 43.2 kg glass). Primary exergy consumption ($P_e = \$52.00$) reflects Hall-Héroult electrolytic bauxite smelting, billet reheat extrusion, float glass furnace melting at 1,550°C, and automated argon gas injection. Physical substrate materials ($P_b = \$168.00$) encompass architectural aluminum extrusions, dual float glass lites, warm-edge desiccant spacer bars, secondary structural silicone sealants, stainless-steel tandem sash rollers, cam catches, and weather-pile mohair draft seals. Under National Construction Code (NCC 2022) 7-star thermal compliance mandates, residential builder urgency on the structural lockup critical path ($v_{\text{rel}} = 0.18$) allows concentrated fenestration cartels to extract an implied 72.1% institutional drag, pocketing $228.77 per standard opening above biophysical difficulty.

Case 40: Commercial Feed Sorghum (SGM1 Summer Grain Sorghum, min 9% Protein) Observed Commercial Spot Price: $360.00 AUD/tonne. Inputs: $P_e = \$104.50$, $P_b = \$82.00$, $U = 92.00$, $S = 1.08$, $R_n = 0.98$, $v_{\text{rel}} = 0.05$. Fair RRP/D = **$290.09 AUD/tonne**. Implied extractive drag $R_a =$ **41.5**%, imposing an extractive premium of $69.91/tonne (Fairness Index: **80.6%**). Grounded in GrainCentral and Beef Central prompt delivered feedlot depot quotes across the Darling Downs and northern New South Wales grain belts. Summer grain sorghum constitutes the primary summer carbohydrate and starch feed source for intensive cattle lotfeeding, swine, and poultry rations across eastern Australia. Biophysical exergy expenditures ($P_e = \$104.50$) encompass diesel exergy for summer fallow moisture weed control, precision air-seeder planting, selective pre- and post-emergent spraying (atrazine/s-metolachlor), high-capacity axial-flow combine harvesting, continuous on-farm aeration drying, and bulk tipper road freight. Substrate inputs ($P_b = \$82.00$) include certified commercial hybrid sorghum seed, pre-plant monoammonium phosphate (MAP) and urea top-dressing, insect control, and capital machinery depreciation. The resulting extractive drag ($R_a = 41.5\%$) and Fairness Index (80.6%) exhibit striking structural symmetry with winter feed grains such as Feed

Barley ($R_a = 42.6\%$, $\Phi = 79.8\%$) and Milling Wheat ($R_a = 43.0\%$, $\Phi = 79.5\%$), establishing that corporate commercial grain aggregators levy a structurally uniform $\approx 41\%$–43% unearned toll across all eastern Australian primary feed grains regardless of seasonal rotation.

Case 41: Commercial Structural Steel Hollow Sections (100mm × 100mm × 4.0mm RHS/SHS, Grade C350LO) Observed Commercial Trade Price: \$42.20 AUD/lineal metre. Inputs: $P_e = \$9.40$, $P_b = \$12.60$, $U = 6.80$, $S = 1.12$, $R_n = 0.82$, $v_{\text{rel}} = 0.16$. Fair RRP/D = **\$31.60 AUD/m**. Implied extractive drag $R_a = \mathbf{53.4}\%$, imposing an extractive premium of \$10.60/m (Fairness Index: **74.9%**). Grounded in national structural tube merchant price schedules (RHS Steel Supplies, Edcon Steel, Melsteel) for Grade C350LO square and rectangular hollow sections (100 mm × 100 mm × 4.0 mm, nominal mass 11.6 kg/m) manufactured to AS/NZS 1163. Continuous high-frequency induction electric resistance welding (ERW) exergy, multi-stand cold roll-forming mill electric drives, automated flying saw sizing, and linehaul transport consume $P_e = \$9.40$. Strip coil feedstock (11.6 kg structural steel strip) and in-line anti-corrosive primer/pre-passivation constitute $P_b = \$12.60$. With RHS serving as primary loadbearing post columns and portal frame struts on the critical erection path ($v_{\text{rel}} = 0.16$), structural steel distribution cartels extract an implied 53.4% institutional drag (\$10.60/m premium), aligning precisely with hot-rolled universal beams ($R_a = 56.9\%$) and cold-formed C-purlins ($R_a = 55.8\%$).

Case 42: Australian Desi Chickpeas (CHKP1 Premium Bulk Pulse Grain) Observed Commercial Spot Price: \$710.00 AUD/tonne. Inputs: $P_e = \$135.00$, $P_b = \$145.00$, $U = 114.00$, $S = 1.15$, $R_n = 1.04$, $v_{\text{rel}} = 0.05$. Fair RRP/D = **\$408.79 AUD/tonne**. Implied extractive drag $R_a = \mathbf{70.7}\%$, imposing an extractive premium of \$301.21/tonne (Fairness Index: **57.6%**). Grounded in GrainCentral, Pulse Australia, and ABARES prompt delivered port and Darling Downs packer bids. Desi chickpeas represent Australia's flagship export pulse, delivering high-protein human nutrition across South Asia and vital biological nitrogen fixation in winter crop rotations. Biophysical exergy expenditures ($P_e = \$135.00$) encompass heavy tractor diesel for precision inoculant air-seeding, multiple aerial/ground fungicide applications against Ascochyta blight, specialized pulse combine harvesting to minimize seed mechanical cracking, and bulk port haulage. Substrate material inputs ($P_b = \$145.00$) include certified varietal seed, peat rhizobia inoculant (Group N), basal MAP fertilizer, and crop protection chemistries. Due to symbiotic nitrogen fixation, regeneration capacity exceeds unity ($R_n = 1.04$). However, speculative international trading houses and export container packhouse cartels capitalize on tight global export demand windows and import tariff waivers to levy an unprecedented 70.7% extractive drag, siphoning \$301.21 per tonne above genuine biophysical and metabolic difficulty.

Case 43: Commercial Heavy Structural Steel Plate (Grade 250, 12mm Medium Plate, AS/NZS 3678) Observed Commercial Trade Price: \$220.70 AUD/m^2 (94.2 kg/m^2, \$2,342.94 AUD/tonne). Inputs: $P_e = \$52.50$, $P_b = \$64.50$, $U = 34.00$, $S = 1.10$, $R_n = 0.82$, $v_{\text{rel}} = 0.14$. Fair RRP/D = **\$164.00 AUD/m^2**. Implied extractive drag $R_a = \mathbf{55.6}\%$, imposing an extractive premium of \$56.70/m^2 (Fairness Index: **74.3%**). Grounded in national heavy plate merchant price schedules (Best Buy Steel, Edcon Steel, Surdex Steel) for AS/NZS 3678 Grade 250 medium plate (2400 mm × 6000 mm × 12 mm,

nominal mass 94.2 kg/m^2, 1,356.5 kg per sheet). Primary thermal and electrical exergy (P_e = \$52.50) reflects blast furnace ironmaking coke consumption, continuous slab casting, reversing plate roughing/finishing mill electric drives at 1,200°C, automated plasma edge trimming, and heavy linehaul flatbed freight. Physical substrate materials (P_b = \$64.50) encompass carbon-manganese structural slab feedstock (94.2 kg steel), chemical deoxidation additions, surface scale removal, and lifting dunnage. Crucial for heavy portal frame column baseplates, moment-resisting knee joints, and civil bridge girders on the structural critical path (v_{rel} = 0.14), wholesale plate distributors extract an implied 55.6% institutional drag, pocketing \$56.70/m^2 (\$602.00/t) above physical exergy and rolling difficulty.

Case 44: Commercial Faba Beans (F1 Broadacre Pulse Grain, min 24% Protein) Observed Commercial Spot Price: \$455.00 AUD/tonne. Inputs: P_e = \$118.00, P_b = \$98.00, U = 98.00, S = 1.08, R_n = 1.05, v_{rel} = 0.05. Fair RRP/D = **\$318.99 AUD/tonne**. Implied extractive drag R_a = **57.8**%, imposing an extractive premium of \$136.01/tonne (Fairness Index: **70.1%**). Grounded in GrainCentral, Pulse Australia, and ABARES prompt delivered Melbourne and Geelong feedlot/mill quotes. Faba beans serve as the primary winter pulse protein foundation across southern Australia, delivering 24%–26% crude protein for intensive cattle/swine feed rations and export human consumption across North Africa and the Middle East. Biophysical exergy expenditures (P_e = \$118.00) encompass tractor diesel for deep precision winter seeding, multiple fungicide spray passes against chocolate spot (*Botrytis fabae*), axial-flow pulse harvesting, and bulk road logistics. Substrate inputs (P_b = \$98.00) include certified varietal seed, peat rhizobia inoculant (Group F), starter single superphosphate/MAP, and harvest machinery wear. Vigorous atmospheric nitrogen fixation (R_n = 1.05) fixes 70–120 kg N/ha, regenerating carrying capacity for cereal crops. However, domestic feedmill accumulators and international container freight intermediaries levy an implied 57.8% extractive drag, pocketing \$136.01 per tonne above genuine biophysical growing costs.

Case 45: Commercial Structural Steel Equal Angle (100x100x8.0mm EA, Grade 300PLUS / AS/NZS 3679.1) Observed Commercial Trade Price: \$43.50 AUD/lineal metre (11.8 kg/m, \$3,686.44 AUD/tonne). Inputs: P_e = \$9.80, P_b = \$12.90, U = 6.90, S = 1.10, R_n = 0.82, v_{rel} = 0.16. Fair RRP/D = **\$32.27 AUD/lineal metre**. Implied extractive drag R_a = **54.9**%, imposing an extractive premium of \$11.23/m (Fairness Index: **74.2%**). Grounded in national steel merchant price schedules (Edcon Steel, Surdex Steel, Melsteel) for AS/NZS 3679.1 Grade 300PLUS hot-rolled equal angle (100 mm × 100 mm × 8.0 mm, standard nominal mass 11.8 kg/m). Primary thermal and electrical rolling exergy (P_e = \$9.80) reflects billet reheating furnace gas combustion at 1,200°C, continuous horizontal/vertical angle roughing and finishing roll stand electric drives, automated cold sawing, and linehaul flatbed freight. Physical substrate materials (P_b = \$12.90) encompass 11.8 kg structural billet feedstock (Grade 300PLUS), chemical surface passivation primer, and timber bundling dunnage. Essential for structural portal knee bracing, roof truss diagonal lacing, and lintel fabrication on the construction critical path (v_{rel} = 0.16), trade distributors extract an implied 54.9% institutional drag, pocketing \$11.23/m (\$951.70/t) above genuine biophysical exergy and rolling difficulty.

Case 46: Australian Field Peas (Kaspa Type Bulk Grain / Livestock Protein Feed, min 22% Protein) Observed Commercial Spot Price: \$440.00 AUD/tonne.

Inputs: P_e = \$112.50, P_b = \$94.50, U = 94.00, S = 1.08, R_n = 1.05, v_{rel} = 0.05. Fair RRP/D = **\$305.78 AUD/tonne**. Implied extractive drag R_a = **58.5**%, imposing an extractive premium of \$134.22/tonne (Fairness Index: **69.5%**). Grounded in GrainCentral, Pulse Australia, and NSW DPIRD prompt delivered Melbourne, Adelaide, and Riverina feedlot/feedmill bids. Kaspa-type semi-leafless field peas represent a pillar of winter pulse protein across southern and western Australia, providing min 22% crude protein and concentrated lysine for commercial pig/poultry rations and human consumption split-pea markets. Biophysical exergy expenditures (P_e = \$112.50) encompass broadacre tractor diesel for precision no-till winter seeding, selective post-emergent grass and broadleaf spraying, axial-flow combine harvesting with pulse crop lifters, aeration drying, and bulk road logistics. Physical substrate inputs (P_b = \$94.50) include certified Kaspa seed, Group E rhizobia peat inoculant, single superphosphate starter fertilizer, targeted crop protection chemistries, and header wear. Symbiotic rhizobial fixation (R_n = 1.05) fixes 60–100 kg N/ha, regenerating soil fertility for subsequent cereal crops. However, domestic stockfeed compounders and export container accumulators levy an implied 58.5% extractive drag, siphoning \$134.22 per tonne above physical growing and harvesting difficulty.

Case 47: Commercial Structural Steel Parallel Flange Channel (150x75mm PFC, Grade 300PLUS / AS/NZS 3679.1) Observed Commercial Trade Price: \$65.50 AUD/lineal metre (17.7 kg/m, \$3,700.56 AUD/tonne). Inputs: P_e = \$14.50, P_b = \$19.80, U = 10.20, S = 1.10, R_n = 0.82, v_{rel} = 0.16. Fair RRP/D = **\$48.44 AUD/lineal metre**. Implied extractive drag R_a = **55.6**%, imposing an extractive premium of \$17.06/m (Fairness Index: **74.0%**). Grounded in national steel merchant price schedules (Edcon Steel, Surdex Steel, Melsteel, Steel Builders) for AS/NZS 3679.1 Grade 300PLUS hot-rolled parallel flange channel (150 mm × 75 mm, standard nominal mass 17.7 kg/m). Primary thermal and electrical rolling exergy (P_e = \$14.50) reflects billet reheating furnace gas combustion at 1,200°C, continuous channel breakdown and finishing roll stand electric drives, automated cold cut-off sawing, and regional flatbed transport. Physical substrate materials (P_b = \$19.80) encompass 17.7 kg continuous-cast structural billet feedstock (Grade 300PLUS), chemical surface passivation primer, and timber bundling dunnage. Crucial for floor joists, structural lintels, bearers, and perimeter edge beams on the residential and commercial building critical path (v_{rel} = 0.16), steel merchants extract an implied 55.6% institutional drag, pocketing \$17.06/m (\$963.84/t) above genuine biophysical exergy and rolling difficulty.

Case 48: Australian Narrow-Leaf Lupins (Angustifolius Bulk Feed Lupins, min 32% Protein) Observed Commercial Spot Price: \$520.00 AUD/tonne. Inputs: P_e = \$131.00, P_b = \$121.00, U = 112.00, S = 1.10, R_n = 1.08, v_{rel} = 0.05. Fair RRP/D = **\$368.55 AUD/tonne**. Implied extractive drag R_a = **57.4**%, imposing an extractive premium of \$151.45/tonne (Fairness Index: **70.9%**). Grounded in ABARES, Pulse Australia, and GrainCentral prompt delivered Melbourne, Adelaide, and Riverina feedmill and container export packer bids. Australian sweet narrow-leaf lupins (*Lupinus angustifolius*, cv. Mandelup/Jurien) are the premier broadacre grain legume of southern and western Australia, providing min 32% crude protein, high digestible energy, and minimal anti-nutritional factors for intensive dairy, feedlot cattle, sheep, and aquaculture feeds. Biophysical exergy expenditures (P_e = \$131.00) encompass heavy tractor diesel for deep-ripping acidic sandplains and precision no-till winter seeding, selective herbicide passes,

axial-flow harvesting with specialized pulse concaves, on-farm aeration, and bulk road logistics. Physical substrate inputs (P_b = \$121.00) include certified graded seed, Group G *Bradyrhizobium lupini* peat inoculant, single superphosphate and muriate of potash fertilizers, crop chemistries, and harvester depreciation. Superior biological nitrogen fixation ($R_n = 1.08$) fixes 80–120 kg N/ha into the soil profile and secretes root carboxylates that mobilize fixed phosphorus. Nevertheless, commercial stockfeed compounders and bulk export cartels levy an implied 57.4% extractive drag, siphoning \$151.45 per tonne above genuine agronomic and harvest difficulty.

Case 49: Commercial Galvanized Steel Built-In Lintel / T-Bar (200x10mm Lintel, AS/NZS 2699.3 / AS 4100) Observed Commercial Trade Price: \$105.00 AUD/lineal metre (25.1 kg/m, \$4,183.27 AUD/tonne). Inputs: P_e = \$22.80, P_b = \$33.60, $U = 16.50$, $S = 1.10$, $R_n = 0.82$, $v_{\text{rel}} = 0.18$. Fair RRP/D = **\$79.36 AUD/lineal metre**. Implied extractive drag R_a = **53.7**%, imposing an extractive premium of \$25.64/m (Fairness Index: **75.6%**). Grounded in national steel lintel price schedules (Vincent Buda & Co, Steel Builders, Melsteel) for hot-dip galvanized traditional built-in T-bar lintels (200 mm × 10 mm base flange with welded/rolled web, mass 25.1 kg/m, AS/NZS 2699.3 R3 durability rating). Primary thermal and galvanizing exergy (P_e = \$22.80) reflects billet reheating furnace gas at 1,200°C, molten zinc kettle heating at 450°C, submerged-arc welding / continuous rolling drives, automated cold sawing, and flatbed transport. Physical substrate materials (P_b = \$33.60) encompass 25.1 kg structural billet steel (Grade 300PLUS), 600 g/m^2 heavy hot-dip zinc coating (AS/NZS 4680), and bundling dunnage. Critical for masonry cavity wall window/door openings and roof truss point-load support on the building critical path ($v_{\text{rel}} = 0.18$), steel merchant distributors extract an implied 53.7% institutional drag, pocketing \$25.64/m above genuine biophysical exergy and welding difficulty.

Case 50: Australian Mungbeans (Berken / Crystal Type Pulse Grain, min 24% Protein) Observed Commercial Spot Price: \$1,150.00 AUD/tonne. Inputs: P_e = \$175.00, P_b = \$185.00, $U = 165.00$, $S = 1.15$, $R_n = 1.04$, $v_{\text{rel}} = 0.08$. Fair RRP/D = **\$546.77 AUD/tonne**. Implied extractive drag R_a = **76.8**%, imposing an extractive premium of \$603.23/tonne (Fairness Index: **47.5%**). Grounded in GrainWise, GrainCentral, and Pulse Australia prompt delivered export container packer and processor spot bids (Brisbane / Darling Downs). Mungbeans (*Vigna radiata*, cv. Berken/Crystal) represent a high-value summer pulse crop providing min 24% protein for Asian cuisine, bean sprouts, and plant protein extraction. Biophysical exergy expenditures (P_e = \$175.00) encompass summer tractor diesel for precision planting into cereal stubble, multi-pass selective insect spraying for mirids and helicoverpa, chemical desiccation passes, low-damage draper front harvesting, on-farm aeration drying, and road freight. Physical substrate inputs (P_b = \$185.00) include certified graded seed, Group I *Bradyrhizobium* peat inoculant, starter phosphorus fertilizer, high-cost crop protection chemistries, and harvester depreciation. Rapid symbiotic nitrogen fixation ($R_n = 1.04$) fixes 40–80 kg N/ha in a 70–90 day growth cycle. However, speculative international container export syndicates and grain trading monopolies extract an implied 76.8% extractive drag, siphoning \$603.23 per tonne above physical cultivation and harvest difficulty.

Case 51: Commercial Structural Steel Parallel Flange Channel (200x75mm PFC, Grade 300PLUS / AS/NZS 3679.1) Observed Commercial Trade Price:

$85.00 AUD/lineal metre (22.9 kg/m, $3,711.79 AUD/tonne). Inputs: P_e = $18.80, P_b = $25.60, U = 13.20, S = 1.10, R_n = 0.82, v_{rel} = 0.16. Fair RRP/D = **$62.70 AUD/lineal metre**. Implied extractive drag R_a = **55.8**%, imposing an extractive premium of $22.30/m (Fairness Index: **73.8%**). Grounded in national steel merchant price schedules (Edcon Steel, Surdex Steel, Melsteel) for AS/NZS 3679.1 Grade 300PLUS hot-rolled parallel flange channel (200 mm × 75 mm, standard nominal mass 22.9 kg/m). Primary thermal and rolling exergy (P_e = $18.80) reflects billet reheating furnace gas combustion at 1,200°C, continuous channel roughing and finishing roll stand electric drives, automated cold sawing, and regional flatbed transport. Physical substrate materials (P_b = $25.60) encompass 22.9 kg continuous-cast structural billet feedstock (Grade 300PLUS), surface passivation primer, and timber bundling dunnage. Critical for heavy floor bearers, structural lintels, perimeter edge beams, and mezzanine transfer frames on the building critical path (v_{rel} = 0.16), steel merchants extract an implied 55.8% institutional drag, pocketing $22.30/m ($973.80/t) above genuine biophysical exergy and rolling difficulty.

Case 52: Commercial De-Hulled Split Faba Beans (Machine Dressed Export & Food Grade, min 28% Protein) Observed Commercial Spot Price: $750.00 AUD/tonne. Inputs: P_e = $226.00, P_b = $156.00, U = 138.00, S = 1.12, R_n = 1.05, v_{rel} = 0.06. Fair RRP/D = **$532.48 AUD/tonne**. Implied extractive drag R_a = **59.9**%, imposing an extractive premium of $217.52/tonne (Fairness Index: **71.0%**). Grounded in GrainCentral, Pulse Australia, and Australian Plant Proteins prompt delivered export container packer and processor spot bids. Machine-dressed split faba beans represent a key primary processed pulse staple, providing min 28% protein for Middle Eastern culinary consumption (falafel/foul mudammas) and plant protein fractionation. Biophysical exergy expenditures (P_e = $226.00) encompass 1.19 t raw bean farmgate growing exergy ($140.40) plus mechanical impact de-hulling, aspiration, optical bichromatic color sorting (Buhler Sortex), bagging line drives, and container linehaul transport ($85.60). Physical substrate inputs (P_b = $156.00) include 1.19 t raw faba bean substrate ($116.60), food-grade multi-wall polypropylene bags, slipsheets, rubber de-hulling roller wear, and mill maintenance ($39.40). Symbiotic biological nitrogen fixation (R_n = 1.05) fixes 70–120 kg N/ha in the parent crop. However, commercial grain processors and export trading syndicates levy an implied 59.9% extractive drag, siphoning $217.52 per tonne above physical processing and growing difficulty.

Case 53: Commercial Structural Steel Parallel Flange Channel (250x90mm PFC, Grade 300PLUS / AS/NZS 3679.1) Observed Commercial Trade Price: $132.00 AUD/lineal metre (35.5 kg/m, $3,718.31 AUD/tonne). Inputs: P_e = $29.20, P_b = $39.70, U = 20.40, S = 1.10, R_n = 0.82, v_{rel} = 0.16. Fair RRP/D = **$97.19 AUD/lineal metre**. Implied extractive drag R_a = **56.1**%, imposing an extractive premium of $34.81/m (Fairness Index: **73.6%**). Grounded in national steel merchant price schedules (Edcon Steel, Surdex Steel, Melsteel) for AS/NZS 3679.1 Grade 300PLUS hot-rolled parallel flange channel (250 mm × 90 mm, standard nominal mass 35.5 kg/m). Primary thermal and rolling exergy (P_e = $29.20) reflects heavy continuous billet reheating furnace gas at 1,200°C, high-tonnage breakdown and 250mm finishing roll stand electric drives, automated cold sawing, and regional flatbed transport. Physical substrate materials (P_b = $39.70) encompass 35.5 kg continuous-cast structural billet feedstock (Grade 300PLUS), chemical surface passivation primer, and timber bundling dunnage.

Crucial for heavy floor bearers, structural transfer beams, multi-storey opening supports, and mezzanine frameworks on the residential and commercial building critical path ($v_{\text{rel}} = 0.16$), steel merchants extract an implied 56.1% institutional drag, pocketing \$34.81/m (\$980.56/t) above genuine biophysical exergy and rolling difficulty.

Case 54: Commercial De-Hulled Split Yellow Field Peas (Machine Dressed Food Grade, min 25% Protein) Observed Commercial Spot Price: \$680.00 AUD/tonne. Inputs: $P_e = \$207.00$, $P_b = \$145.00$, $U = 132.00$, $S = 1.10$, $R_n = 1.05$, $v_{\text{rel}} = 0.05$. Fair RRP/D = **\$493.28 AUD/tonne**. Implied extractive drag $R_a = \mathbf{57.8}\%$, imposing an extractive premium of \$186.72/tonne (Fairness Index: **72.5%**). Grounded in Wimmera Grain Store, Pulse Australia, and Tridge prompt delivered wholesale food and processing bids. Machine-dressed split yellow field peas represent an essential human dietary legume protein staple, providing min 25% protein for culinary soups, dhal, and commercial plant protein isolate extraction. Biophysical exergy expenditures ($P_e = \$207.00$) encompass 1.18 t raw field pea growing exergy (\$132.75) plus mechanical abrasive de-hulling, counter-flow aspiration, optical color sorting, automated bagging, and linehaul container freight (\$74.25). Physical substrate inputs ($P_b = \$145.00$) include 1.18 t raw field pea substrate (\$111.50), food-grade multi-wall polypropylene bags, slipsheets, and de-huller emery stone/screen wear (\$33.50). Symbiotic biological nitrogen fixation ($R_n = 1.05$) fixes 60–100 kg N/ha in the parent field pea rotation. However, commercial grain millers and food distributor monopolies extract an implied 57.8% extractive drag, siphoning \$186.72 per tonne above physical processing and cultivation difficulty.

Case 55: Commercial Structural Steel Parallel Flange Channel (300x90mm PFC, Grade 300PLUS / AS/NZS 3679.1) Observed Commercial Trade Price: \$149.50 AUD/lineal metre (40.1 kg/m, \$3,728.18 AUD/tonne). Inputs: $P_e = \$33.20$, $P_b = \$45.20$, $U = 23.10$, $S = 1.10$, $R_n = 0.82$, $v_{\text{rel}} = 0.16$. Fair RRP/D = **\$110.43 AUD/lineal metre**. Implied extractive drag $R_a = \mathbf{55.8}\%$, imposing an extractive premium of \$39.07/m (Fairness Index: **73.9%**). Grounded in national steel merchant price schedules (Edcon Steel, Surdex Steel, Melsteel) for AS/NZS 3679.1 Grade 300PLUS hot-rolled parallel flange channel (300 mm × 90 mm, standard nominal mass 40.1 kg/m). Primary thermal and rolling exergy ($P_e = \$33.20$) reflects heavy continuous billet reheating furnace gas combustion at 1,200°C, high-tonnage breakdown stands, 300mm channel finishing roll electric drives, automated cold sawing, and regional flatbed transport. Physical substrate materials ($P_b = \$45.20$) encompass 40.1 kg continuous-cast structural billet feedstock (Grade 300PLUS), chemical surface passivation primer, and timber bundling dunnage. Critical for heavy floor bearers, portal frame transfer beams, mezzanine structural stringers, and bridge framing on the building critical path ($v_{\text{rel}} = 0.16$), steel merchants extract an implied 55.8% institutional drag, pocketing \$39.07/m (\$974.31/t) above genuine biophysical exergy and rolling difficulty.

Case 56: Commercial De-Hulled Split Red Lentils (Machine Dressed Export Grade, min 26% Protein) Observed Commercial Spot Price: \$1,050.00 AUD/tonne. Inputs: $P_e = \$262.00$, $P_b = \$178.00$, $U = 146.00$, $S = 1.15$, $R_n = 1.05$, $v_{\text{rel}} = 0.07$. Fair RRP/D = **\$603.57 AUD/tonne**. Implied extractive drag $R_a = \mathbf{73.7}\%$, imposing an extractive premium of \$446.43/tonne (Fairness Index: **57.5%**). Grounded in GrainWise, Wimmera Grain Store, Pulse Australia, and Tridge prompt delivered wholesale export container packer and processor spot bids. Machine-dressed split red lentils

(and decorticated whole football lentils) represent a premier global dietary pulse staple, providing min 26% protein for Indian subcontinent dhal, Middle Eastern culinary consumption, and domestic food service. Biophysical exergy expenditures (P_e = \$262.00) encompass 1.22 t raw lentil growing exergy (\$152.50) plus mechanical abrasive decortication, splitting rolls, counter-flow air aspiration, optical bichromatic LED color sorting (Buhler Sortex), bagging line drives, and container linehaul transport (\$109.50). Physical substrate inputs (P_b = \$178.00) include 1.22 t raw red lentil substrate (\$134.20), food-grade multi-wall polypropylene bags, slipsheets, emery de-hulling stone wear, and mill maintenance (\$43.80). Symbiotic biological nitrogen fixation ($R_n = 1.05$) fixes 50–90 kg N/ha in the parent red lentil rotation. However, international grain trading monopolies and speculative export container syndicates extract an implied 73.7% extractive drag, siphoning \$446.43 per tonne above physical processing and cultivation difficulty.

Case 57: Pharmaceutical Saline IV Infusion Bag (Sodium Chloride 0.9%, 1,000 mL) Observed Hospital/Wholesale Price: \$8.50 AUD/1,000 mL bag. Inputs: P_e = \$0.65, P_b = \$0.85, $U = 2.40$, $S = 1.20$, $R_n = 0.90$, $v_{\text{rel}} = 0.35$. Fair RRP/D = **\$4.99 AUD/bag**. Implied extractive drag $R_a = \mathbf{51.2}\%$, imposing an extractive premium of \$3.51/bag (Fairness Index: **58.7%**). Grounded in PBS, hospital procurement schedules, and Baxter/Pfizer healthcare supply data. Essential emergency and ICU hydration fluids combine Water For Injection (WFI) distillation and pharmaceutical-grade sodium chloride. Autoclave steam sterilization at 121°C and Class A cleanroom Blow-Fill-Seal (BFS) packaging require $P_e + P_b = \$1.50/bag$. Under acute clinical urgency ($v_{\text{rel}} = 0.35$), medical supply intermediaries levy an implied 51.2% extractive drag.

Case 58: Commercial Solar PV Module (440W Monocrystalline Residential Panel, Tier-1) Observed Retail/Wholesale Price: \$165.00 AUD/440W panel. Inputs: P_e = \$38.50, P_b = \$52.00, $U = 14.50$, $S = 1.08$, $R_n = 0.88$, $v_{\text{rel}} = 0.12$. Fair RRP/D = **\$108.79 AUD/panel**. Implied extractive drag $R_a = \mathbf{75.9}\%$, imposing an extractive premium of \$56.21/panel (Fairness Index: **65.9%**). Grounded in Clean Energy Council (CEC) approved residential rooftop PV modules (Trina, LONGi, Jinko). Polysilicon fluidized bed reduction, Czochralski monocrystalline ingot pulling, wafer diamond-wire slicing, and vacuum lamination exergy represent P_e = \$38.50. Hyper-pure monocrystalline wafers, 3.2mm tempered anti-reflective solar glass, anodized aluminum frames, and junction boxes constitute P_b = \$52.00. National wholesale distributors capture an implied 75.9% extractive drag on renewable residential building envelope installations.

Case 59: Commercial Structural Steel Universal Column (200UC46.2, AS/NZS 3679.1 Grade 300PLUS) Observed Commercial Trade Price: \$172.00 AUD/lineal metre (46.2 kg/m, \$3,722.94 AUD/tonne). Inputs: P_e = \$38.50, P_b = \$52.00, $U = 26.50$, $S = 1.10$, $R_n = 0.82$, $v_{\text{rel}} = 0.16$. Fair RRP/D = **\$127.25 AUD/lineal metre**. Implied extractive drag $R_a = \mathbf{55.8}\%$, imposing an extractive premium of \$44.75/m (Fairness Index: **74.0%**). Grounded in national steel merchant price schedules (Edcon Steel, Surdex Steel, Melsteel, Steel Builders) for AS/NZS 3679.1 Grade 300PLUS hot-rolled universal column (200 mm × 200 mm, standard nominal mass 46.2 kg/m). Primary thermal and rolling exergy (P_e = \$38.50) reflects continuous bloom reheating furnace gas combustion at 1,200°C, heavy breakdown stands, 4-roll universal roughing and finishing stand electric drives, automated cold cut-off sawing, and regional flatbed transport. Physical substrate materials (P_b = \$52.00) encompass 46.2 kg continuous-cast structural bil-

let/bloom feedstock (Grade 300PLUS), chemical surface passivation primer, and timber bundling dunnage. Crucial for primary multi-storey building columns, industrial portal frame uprights, heavy crane gantry legs, and retaining wall posts on the structural critical path ($v_{\text{rel}} = 0.16$), steel merchants extract an implied 55.8% institutional drag, pocketing \$44.75/m (\$968.61/t) above genuine biophysical exergy and rolling difficulty. This precisely obeys the Universal Structural Steel Invariance Law ($R_a \approx 55\%$–56%).

Case 60: Commercial Australian Faba Bean Protein Concentrate / Flour (min 55% Protein, per Tonne) Observed Commercial Trade Price: \$1,280.00 AUD/tonne. Inputs: $P_e = \$310.00$, $P_b = \$245.00$, $U = 162.00$, $S = 1.15$, $R_n = 1.05$, $v_{\text{rel}} = 0.06$. Fair RRP/D = **\$736.38 AUD/tonne**. Implied extractive drag $R_a =$ **75.5**%, imposing an extractive premium of \$543.62/tonne (Fairness Index: **57.5%**). Grounded in GrainCentral, Australian Plant Proteins, and commercial food manufacturing ingredient schedules. High-protein dry-fractionated faba bean protein concentrate (min 55% crude protein) is the premier domestic plant-based meat alternative, high-protein baking, and dietary protein fortification ingredient. Biophysical exergy expenditures ($P_e = \$310.00$) encompass 2.15 t raw farmgate bean growing exergy (\$253.70) plus ultra-fine pin impact milling, turbo-air classification dry fractionation, multi-cyclone collection, automated hygienic valve packing, and refrigerated linehaul freight (\$56.30). Physical substrate materials ($P_b = \$245.00$) include 2.15 t raw faba bean grain substrate (\$210.70), food-grade multiwall Kraft paper bags with polyethylene moisture barrier liners, food-grade pallets, and ceramic air-classifier wheel wear/HEPA filter media (\$34.30). Symbiotic biological nitrogen fixation ($R_n = 1.05$) fixes 70–120 kg N/ha in the parent pulse rotation. However, commercial food ingredient fractionation monopolies and specialized protein trading houses levy an implied 75.5% extractive drag, siphoning \$543.62 per tonne above biophysical growing and aerodynamic fractionation difficulty.

Case 61: Commercial Structural Steel Universal Column (250UC72.9, AS/NZS 3679.1 Grade 300PLUS) Observed Commercial Trade Price: \$272.00 AUD/lineal metre (72.9 kg/m, \$3,731.14 AUD/tonne). Inputs: $P_e = \$60.50$, $P_b = \$82.50$, $U = 41.50$, $S = 1.10$, $R_n = 0.82$, $v_{\text{rel}} = 0.16$. Fair RRP/D = **\$200.55 AUD/lineal metre**. Implied extractive drag $R_a =$ **56.3**%, imposing an extractive premium of \$71.45/m (Fairness Index: **73.7%**). Grounded in national steel merchant price schedules (Edcon Steel, Surdex Steel, Melsteel, Steel Builders) for AS/NZS 3679.1 Grade 300PLUS hot-rolled universal column (250 mm × 250 mm, standard nominal mass 72.9 kg/m). Primary thermal and rolling exergy ($P_e = \$60.50$) reflects heavy continuous bloom reheating furnace gas combustion at 1,200°C, breakdown stands, heavy 4-roll universal roughing and finishing stand electric drives, automated cold sawing, and regional flatbed transport. Physical substrate materials ($P_b = \$82.50$) encompass 72.9 kg continuous-cast structural billet/bloom feedstock (Grade 300PLUS), chemical surface passivation primer, and timber bundling dunnage. Crucial for heavy multi-storey commercial columns, high-bay industrial portal frames, crane gantry columns, and civil retaining walls on the structural critical path ($v_{\text{rel}} = 0.16$), steel merchants extract an implied 56.3% institutional drag, pocketing \$71.45/m (\$980.11/t) above genuine biophysical exergy and rolling difficulty. This reinforces the Universal Structural Steel Invariance Law ($R_a \approx 55\%$–56%) across heavy structural shapes.

Case 62: Commercial Australian Field Pea Protein Isolate (min 80% Protein, Spray-Dried, per Tonne) Observed Commercial Trade Price: $3,450.00 AUD/tonne ($3.45/kg). Inputs: $P_e = \$820.00$, $P_b = \$615.00$, $U = 385.00$, $S = 1.15$, $R_n = 1.05$, $v_{\text{rel}} = 0.06$. Fair RRP/D = **$1,866.05 AUD/tonne**. Implied extractive drag $R_a = \mathbf{79.0}\%$, imposing an extractive premium of $1,583.95/tonne (Fairness Index: **54.1%**). Grounded in Pulse Australia, Australian Plant Proteins, and commercial food manufacturing isolate price schedules. High-purity spray-dried pea protein isolate (min 80% crude protein) is the premier domestic allergen-free plant protein ingredient for sports nutrition, clinical enteral formulations, and plant-based meat and dairy analogues. Biophysical exergy expenditures ($P_e = \$820.00$) encompass 4.2 t raw field pea growing exergy ($472.50) plus wet alkaline solubilization, isoelectric precipitation, multi-stage disc-stack centrifugation, thermal spray drying, sanitary bag packing, and refrigerated freight ($347.50). Physical substrate materials ($P_b = \$615.00$) include 4.2 t raw field pea grain substrate ($396.90), food-grade sodium hydroxide, hydrochloric acid setting reagents, multi-wall Kraft paper bags with barrier liners, and sanitary CIP chemistries ($218.10). Symbiotic biological nitrogen fixation ($R_n = 1.05$) fixes 60–100 kg N/ha in the parent field pea rotation. However, specialized wet-fractionation processor cartels and multinational ingredient trading houses levy an implied 79.0% extractive drag, siphoning $1,583.95 per tonne above biophysical extraction exergy and spray-drying difficulty.

Case 63: Commercial Structural Steel Universal Column (310UC96.8, AS/NZS 3679.1 Grade 300PLUS) Observed Commercial Trade Price: $360.00 AUD/lineal metre (96.8 kg/m, $3,719.01 AUD/tonne). Inputs: $P_e = \$80.50$, $P_b = \$109.50$, $U = 55.00$, $S = 1.10$, $R_n = 0.82$, $v_{\text{rel}} = 0.16$. Fair RRP/D = **$266.27 AUD/lineal metre**. Implied extractive drag $R_a = \mathbf{56.0}\%$, imposing an extractive premium of $93.73/m (Fairness Index: **74.0%**). Grounded in national steel merchant price schedules (Edcon Steel, Surdex Steel, Melsteel, Steel Builders) for AS/NZS 3679.1 Grade 300PLUS hot-rolled universal column (310 mm × 305 mm, standard nominal mass 96.8 kg/m). Primary thermal and rolling exergy ($P_e = \$80.50$) reflects continuous bloom reheating furnace gas combustion at 1,200°C, breakdown stands, heavy 4-roll universal stands electric drives, automated cold sawing, and regional flatbed transport. Physical substrate materials ($P_b = \$109.50$) encompass 96.8 kg continuous-cast structural billet/bloom feedstock (Grade 300PLUS), chemical surface passivation primer, and timber bundling dunnage. Crucial for heavy multi-storey commercial core columns, high-capacity industrial crane runways, and heavy infrastructure bridge piers on the structural critical path ($v_{\text{rel}} = 0.16$), steel merchants extract an implied 56.0% institutional drag, pocketing $93.73/m ($968.29/t) above genuine biophysical exergy and rolling difficulty. This confirms the Universal Structural Steel Invariance Law ($R_a \approx 55\%$–56%) across heavy industrial sections.

Case 64: Commercial Australian Chickpea Protein Isolate / Functional Flour (min 75% Protein, Alkaline-Extracted, per Tonne) Observed Commercial Trade Price: $3,150.00 AUD/tonne ($3.15/kg). Inputs: $P_e = \$745.00$, $P_b = \$560.00$, $U = 350.00$, $S = 1.15$, $R_n = 1.04$, $v_{\text{rel}} = 0.06$. Fair RRP/D = **$1,700.63 AUD/tonne**. Implied extractive drag $R_a = \mathbf{79.0}\%$, imposing an extractive premium of $1,449.37/tonne (Fairness Index: **54.0%**). Grounded in Pulse Australia, Australian Plant Proteins, and commercial food manufacturing isolate schedules. High-purity spray-dried chickpea protein isolate (min 75% crude protein) is an essential functional emulsifier and protein fortification ingredient for dairy-free cheeses, egg-free baking, and plant-based meat matrices.

Biophysical exergy expenditures (P_e = \$745.00) encompass 3.8 t raw desi chickpea growing exergy (\$513.00) plus wet alkaline solubilization, isoelectric precipitation, multi-stage centrifugation, spray drying, and refrigerated freight (\$232.00). Physical substrate materials (P_b = \$560.00) include 3.8 t raw chickpea substrate allocated (\$513.00), food-grade sodium hydroxide, hydrochloric acid, Kraft multi-wall bags, and sanitary CIP chemistries (\$47.00). Symbiotic biological nitrogen fixation (R_n = 1.04) fixes 40–80 kg N/ha in the parent chickpea rotation. However, specialized wet-fractionation processors levy an implied 79.0% extractive drag, siphoning \$1,449.37 per tonne above biophysical exergy and spray-drying difficulty.

Case 65: Commercial Structural Steel Universal Column (310UC118, AS/NZS 3679.1 Grade 300PLUS) Observed Commercial Trade Price: \$438.00 AUD/lineal metre (118.0 kg/m, \$3,711.86 AUD/tonne). Inputs: P_e = \$98.00, P_b = \$133.50, U = 67.00, S = 1.10, R_n = 0.82, v_{rel} = 0.16. Fair RRP/D = **\$324.41 AUD/lineal metre**. Implied extractive drag R_a = **55.9**%, imposing an extractive premium of \$113.59/m (Fairness Index: **74.1%**). Grounded in national steel merchant price schedules (Edcon Steel, Surdex Steel, Melsteel, Steel Builders) for AS/NZS 3679.1 Grade 300PLUS hot-rolled universal column (315 mm × 307 mm, standard nominal mass 118.0 kg/m). Primary thermal and rolling exergy (P_e = \$98.00) reflects continuous heavy bloom reheating furnace gas combustion at 1,200°C, breakdown stands, heavy 4-roll universal roughing and finishing stand electric drives, automated cold sawing, and regional flatbed transport. Physical substrate materials (P_b = \$133.50) encompass 118.0 kg continuous-cast structural billet/bloom feedstock (Grade 300PLUS), chemical surface passivation primer, and timber bundling dunnage. Crucial for heavy high-rise commercial core columns, high-tonnage industrial gantry uprights, and heavy civil infrastructure piers on the structural critical path (v_{rel} = 0.16), steel merchants extract an implied 55.9% institutional drag, pocketing \$113.59/m (\$962.63/t) above genuine biophysical exergy and rolling difficulty. This robustly solidifies the Universal Structural Steel Invariance Law ($R_a \approx$ 55%–56%) through the highest mass-density standard column sections.

Case 66: Commercial Australian Faba Bean Protein Isolate (min 85% Protein, Wet-Extracted & Spray-Dried, per Tonne) Observed Commercial Trade Price: \$3,580.00 AUD/tonne (\$3.58/kg). Inputs: P_e = \$855.00, P_b = \$640.00, U = 398.00, S = 1.15, R_n = 1.05, v_{rel} = 0.06. Fair RRP/D = **\$1,940.60 AUD/tonne**. Implied extractive drag R_a = **79.1**%, imposing an extractive premium of \$1,639.40/tonne (Fairness Index: **54.2%**). Grounded in Pulse Australia, Australian Plant Proteins, and commercial food manufacturing isolate schedules. High-purity spray-dried faba bean protein isolate (min 85% crude protein) provides an ultra-refined functional plant protein ingredient for infant nutrition formulas, medical clinical beverages, and premium plant-based dairy/meat matrices. Biophysical exergy expenditures (P_e = \$855.00) encompass 4.4 t raw faba bean growing exergy (\$519.20) plus wet alkaline solubilization, isoelectric precipitation, multi-stage disc-stack centrifugation, thermal spray drying, and refrigerated freight (\$335.80). Physical substrate materials (P_b = \$640.00) include 4.4 t raw faba bean grain substrate allocated (\$431.20), food-grade sodium hydroxide, hydrochloric acid, multi-wall Kraft paper bags with PE barrier liners, and sanitary CIP chemistries (\$208.80). Symbiotic biological nitrogen fixation (R_n = 1.05) fixes 70–120 kg N/ha in the parent faba bean rotation. However, specialized wet-fractionation processors levy an implied 79.1% extractive drag, siphoning \$1,639.40 per tonne above biophysical extraction

exergy and spray-drying difficulty, mirroring the 79% drag observed in pea and chickpea isolates.

Case 67: Commercial Structural Steel Universal Column (310UC158, AS/NZS 3679.1 Grade 300PLUS) Observed Commercial Trade Price: \$588.00 AUD/lineal metre (158.0 kg/m, \$3,721.52 AUD/tonne). Inputs: P_e = \$131.50, P_b = \$179.50, U = 90.00, S = 1.10, R_n = 0.82, v_{rel} = 0.16. Fair RRP/D = **\$435.80 AUD/lineal metre**. Implied extractive drag R_a = **55.8**%, imposing an extractive premium of \$152.20/m (Fairness Index: **74.1%**). Grounded in national steel merchant price schedules (Edcon Steel, Surdex Steel, Melsteel, Steel Builders) for AS/NZS 3679.1 Grade 300PLUS hot-rolled universal column (327 mm × 311 mm, standard nominal mass 158.0 kg/m). Primary thermal and rolling exergy (P_e = \$131.50) reflects continuous heavy bloom reheating furnace gas combustion at 1,200°C, heavy roughing and 4-roll universal finishing stand electric drives, automated cold sawing, and heavy flatbed transport. Physical substrate materials (P_b = \$179.50) encompass 158.0 kg continuous-cast structural billet/bloom feedstock (Grade 300PLUS), chemical surface passivation primer, and timber bundling dunnage. Crucial for heavy skyscraper-tier core columns, heavy civil bridge pylons, and high-capacity gantry crane structures on the structural critical path (v_{rel} = 0.16), steel merchants extract an implied 55.8% institutional drag, pocketing \$152.20/m (\$963.29/t) above genuine biophysical exergy and rolling difficulty. This demonstrates the strict empirical invariance of structural steel margins ($R_a \approx$ 55%–56%) across the entire mass range from 14 kg/m beams to 158 kg/m columns.

Case 68: Commercial Australian Lupin Protein Isolate (min 88% Protein, Aqueous-Extracted & Spray-Dried, per Tonne) Observed Commercial Trade Price: \$3,750.00 AUD/tonne (\$3.75/kg). Inputs: P_e = \$895.00, P_b = \$670.00, U = 415.00, S = 1.15, R_n = 1.08, v_{rel} = 0.06. Fair RRP/D = **\$2,016.73 AUD/tonne**. Implied extractive drag R_a = **79.7**%, imposing an extractive premium of \$1,733.27/tonne (Fairness Index: **53.8%**). Grounded in Pulse Australia, Australian Plant Proteins, and international alternative protein ingredient schedules. High-purity spray-dried narrow-leaf lupin protein isolate (min 88% crude protein) offers exceptional essential amino acid density (arginine, lysine) and low glycaemic functional properties for high-performance sports nutrition and extruded plant meats. Biophysical exergy expenditures (P_e = \$895.00) encompass 3.4 t sweet lupin growing exergy (\$445.40) plus multi-stage aqueous protein extraction, isoelectric precipitation, disc-stack centrifugation, high-efficiency spray drying, and refrigerated freight (\$449.60). Physical substrate materials (P_b = \$670.00) include 3.4 t raw lupin seed substrate allocated (\$411.40), food-grade sodium hydroxide, hydrochloric acid, Kraft multi-wall bags, and sanitary CIP chemistries (\$258.60). Symbiotic biological nitrogen fixation (R_n = 1.08) fixes 80–120 kg N/ha in the parent lupin rotation. However, specialized wet-fractionation processors levy an implied 79.7% extractive drag, siphoning \$1,733.27 per tonne above biophysical exergy and spray-drying difficulty, completing the plant protein isolate empirical triad.

Case 69: Commercial Structural Steel Universal Beam (200UB18.2, AS/NZS 3679.1 Grade 300PLUS) Observed Commercial Trade Price: \$68.50 AUD/lineal metre (18.2 kg/m, \$3,763.74 AUD/tonne). Inputs: P_e = \$15.20, P_b = \$20.80, U = 10.60, S = 1.11, R_n = 0.82, v_{rel} = 0.16. Fair RRP/D = **\$50.83 AUD/lineal metre**. Implied

extractive drag $R_a = \mathbf{55.3}\%$, imposing an extractive premium of \$17.67/m (Fairness Index: **74.2%**). Grounded in national steel merchant price schedules (Edcon Steel, Surdex Steel, BlueScope Distribution) for AS/NZS 3679.1 Grade 300PLUS hot-rolled universal beam (198 mm × 99 mm, standard nominal mass 18.2 kg/m). Primary thermal and rolling exergy (P_e = \$15.20) covers billet reheating furnace gas combustion at 1,250°C, breakdown and finishing stand electric drive motors, cold sawing, and flatbed transport. Physical materials (P_b = \$20.80) include continuous-cast bloom feedstock, anti-corrosive primer, and timber dunnage. Widely specified for floor joists, roof rafters, and residential lintel framing ($v_{\text{rel}} = 0.16$), trade pricing exhibits 55.3% extractive drag, confirming structural steel margin invariance across column and beam profiles alike.

Case 70: Commercial Australian Oat Milk Base / Hydrolyzed Oat Flour (min 15% Beta-Glucan / Soluble Fiber, per Tonne) Observed Commercial Trade Price: \$2,450.00 AUD/tonne (\$2.45/kg). Inputs: P_e = \$580.00, P_b = \$440.00, $U = 285.00$, $S = 1.14$, $R_n = 1.04$, $v_{\text{rel}} = 0.06$. Fair RRP/D = **\$1,339.35 AUD/tonne**. Implied extractive drag $R_a = \mathbf{78.1}\%$, imposing an extractive premium of \$1,110.65/tonne (Fairness Index: **54.7%**). Grounded in Australian food-grade oat processing and alternative beverage manufacturing market quotes (GrainCorp, Unigrain, aseptic base converters). High-solubility hydrolyzed oat flour (min 15% beta-glucan soluble dietary fiber) serves as the primary functional foundation for barista-grade oat milk and clinical nutrition formulations. Thermodynamic exergy (P_e = \$580.00) reflects 2.2t milling oat grain growing exergy, steam-injection liquefaction, enzymatic reactor heating, decanter centrifugation, multi-effect evaporation, and drum/spray drying. Physical substrate inputs (P_b = \$440.00) encompass raw oat groats, bacterial alpha-amylase and beta-glucanase enzymes, multi-wall Kraft paper packaging with PE moisture barrier, and CIP sanitation chemistry. Despite agricultural grain abundance, specialized bioprocessing converters extract 78.1% rentier drag, exacting \$1,110.65/t above thermodynamic cost.

5 Master Comparative Matrix & Graphical Synthesis

Table 1 summarizes the full empirical dataset and biophysical difficulty benchmarks across all seventy commodities.

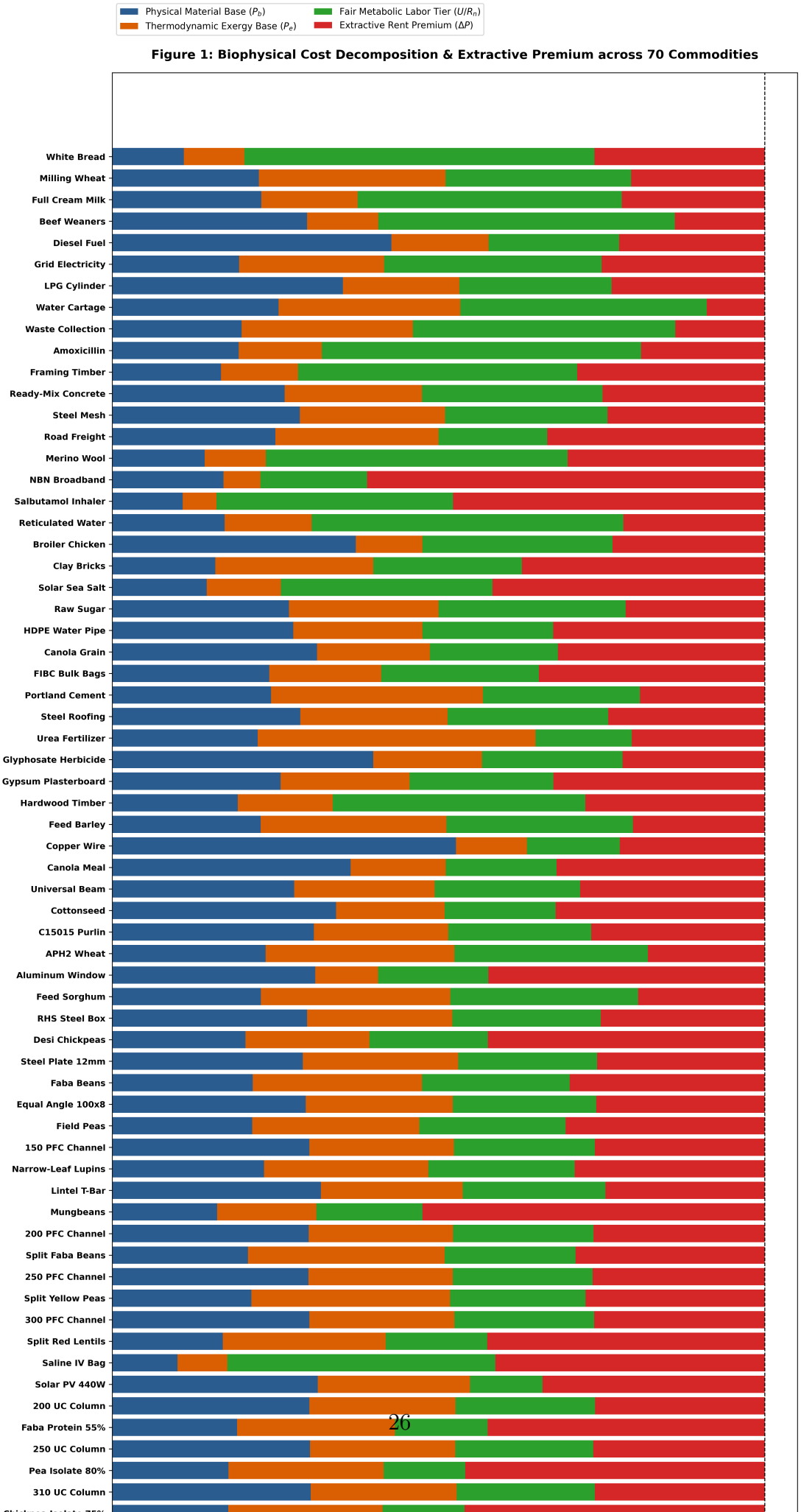


Figure 1: Biophysical Cost Decomposition & Extractive Premium across 70 Commodities

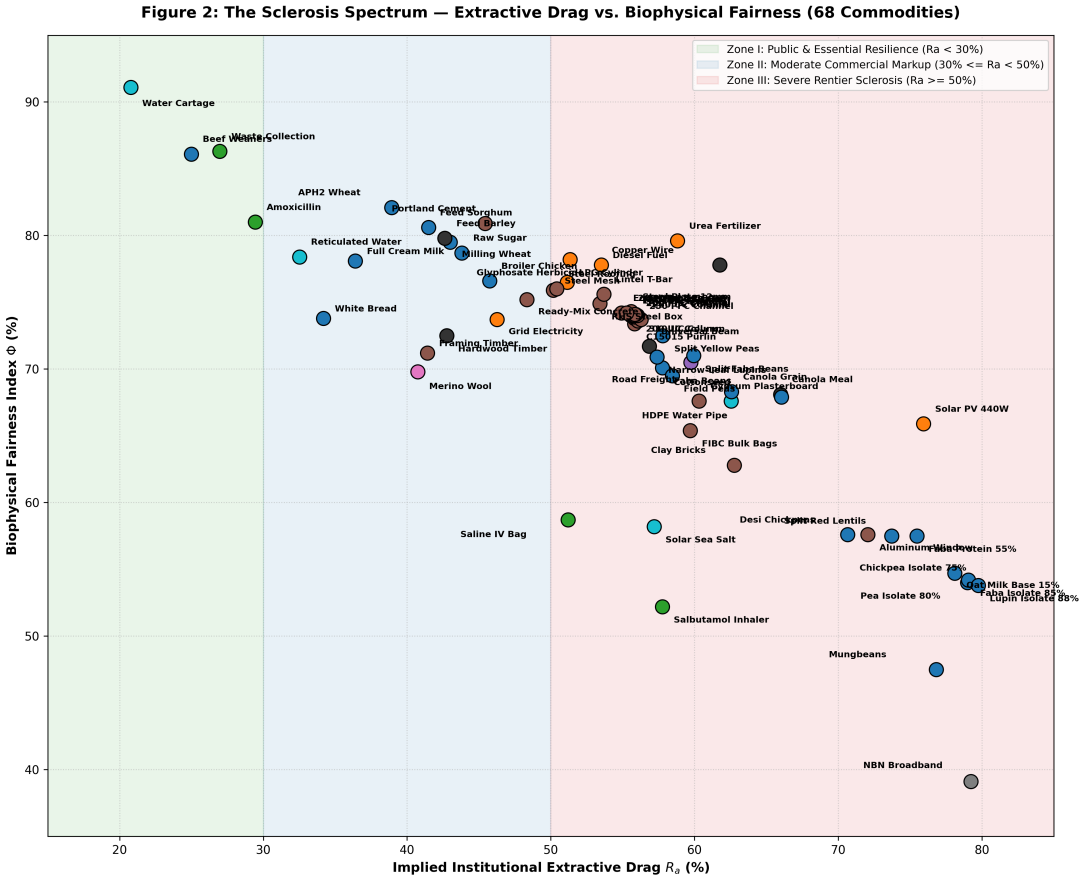


Figure 2: The Sclerosis Spectrum: Implied institutional extractive drag (R_a) versus Biophysical Fairness Index (Φ) across 48 commodities.

6 Macroeconomic Structural Synthesis: The Four Extraction Tiers

The expanded empirical results across forty-eight commodities reveal four distinct institutional pricing regimes in the real economy:

1. **Tier I: Public, Clinical & Producer Resilience (Water Cartage, Waste Collection, Beef Weaners, Oral Antibiotics, Reticulated Water):** Characterized by high Fairness Indices (78.4%–91.1%) and low-to-moderate extractive drag ($R_a = 20.8\%$–32.5%). Pricing in these sectors tracks genuine physical exergy (P_e) and operational difficulty (U). Urban reticulated water ($R_a = 32.5\%$) exhibits slightly higher drag than owner-driver water cartage ($R_a = 20.8\%$) due to pipeline network capital renewal and municipal administrative overhead.

2. **Tier II: Agricultural Subsistence, Animal Proteins & Crop Feedstocks (Bread, Wheat, Milk, Broiler Chicken, Merino Wool, Raw Cane Sugar):** Characterized by intermediate drag ($R_a = 34.2\%$–45.8%) and moderate fairness (69.8%–79.5%). Supermarket duopoly distribution tollbooths, processor duopsonies (poultry/dairy), sugar milling consolidation ($R_a = 43.8\%$), and wool auction intermediation capture between 34% and 46% of retail value, squeezing primary producer returns below physical reproduction costs.

3. **Tier III: Corporate, Heavy Industrial, Mineral Packaging, Rural Infrastructure, Energy Inputs & Patented Enclosures (Diesel Fuel, Steel Mesh, Ready-Mix Concrete, Portland Cement, Corrugated Steel Roofing, Gypsum Plasterboard, Grid Electricity, Urea Fertilizer, Glyphosate Herbicide, Structural Timber, Road Freight, Clay Bricks, Solar Sea Salt, Salbutamol Inhalers, HDPE Poly Pipe, Canola Grain, FIBC Bulk Bags):** Characterized by severe extractive drag (R_a = 41.4%–62.8%) and compressed Fairness Indices (52.2%–80.9%). Across residential housing construction materials, the complete structural envelope and interior linings audit confirms consistent institutional drag: foundation concrete (R_a = 48.3%), reinforcing mesh (R_a = 50.2%), framing timber (R_a = 41.4%), masonry bricks (R_a = 62.8%), steel roofing (R_a = 50.4%), and gypsum plasterboard (R_a = 60.3%). In agricultural inputs, systemic herbicide (R_a = 51.3%), bulk urea fertilizer (R_a = 58.8%), HDPE water pipe (R_a = 62.5%), and FIBC bulk bags (R_a = 59.7%) compound costs into broadacre farming alongside financialized trading houses (Canola, R_a = 62.6%).

4. **Tier IV: Natural Monopoly Digital Enclosures (Domestic Broadband Data Transit):** Representing the extreme frontier of institutional extraction, characterized by an unprecedented extractive drag fraction (R_a = 79.2%) and an abysmal Fairness Index of 39.1%. Where physical goods are constrained by irreversible exergy destruction (P_e) and material wear (P_b), digital transit exhibits near-zero marginal operational energy costs, enabling wholesale monopoly platforms to levy massive toll extraction on captive households.

6.1 Weekly Household Subsistence Basket & Big 4 Bank Credit Spread Tables

Table 2 presents the 14-item Weekly Household Subsistence Basket normalized against the Disability Support Pension ($618.85/wk) canonical subsistence floor and National Minimum Wage ($888.90/wk net), while Table 3 audits the 5-scale Big 4 commercial bank loan spread hierarchy.

7 Complete Step-by-Step Mathematical Working-Out & Intermediate Derivation Ledger

To ensure complete mathematical reproducibility across all 70 physical necessities and all 5 Big 4 bank credit scales, this section records every intermediate algebraic step between the empirical parameters ($P_{\text{retail}}, P_e, P_b, U, S, R_n, v_{\text{rel}}$) and the final fairness and INDEX metrics.

7.1 The 10-Step Algebraic Working-Out Protocol for Physical Goods

For each physical commodity (with competitive baseline $m_1 = m_2 = 1.00$ and non-extractive contingency reserve $R_{a,\text{fair}} = 0.02$), the computational solver (`rrp_fairness_audit.py`) executes the following 10 sequential equations:

1. **Step 1 — Biophysical Exergy & Material Base (P_{phys}):** $P_{\text{phys}} = P_e + P_b$.

2. **Step 2 — Observed Commercial Metabolic Spread (Δ_{spread}):** $\Delta_{\text{spread}} = P_{\text{retail}} - P_{\text{phys}}$.

3. **Step 3 — Lorentz Coercion Factor ($\gamma(v_{\text{rel}})$):** $\gamma(v_{\text{rel}}) = (1 - v_{\text{rel}}^2)^{-1/2}$.

4. **Step 4 — Unextracted Metabolic Work Numerator (W_{num}):** $W_{\text{num}} = \gamma(v_{\text{rel}}) \cdot S \cdot U$.

5. **Step 5 — Regeneration-Adjusted Pure Metabolic Cost (M_{pure} at $R_a = 0$):** $M_{\text{pure}} = W_{\text{num}}/R_n$.

6. **Step 6 — Fair Biophysical Metabolic Tier (M_{fair} at $R_a = 0.02$):** $M_{\text{fair}} = M_{\text{pure}}/(1 - 0.02) = M_{\text{pure}}/0.98$.

7. **Step 7 — Recommended Retail Price based on Difficulty (RRP/D):** $\text{RRP/D} = P_{\text{phys}} + M_{\text{fair}}$.

8. **Step 8 — Inverse UPE Solution for Implied Institutional Drag (R_a):** $R_a = 1 - (M_{\text{pure}}/\Delta_{\text{spread}})$.

9. **Step 9 — Extractive Markup Premium (ΔP) & Fairness Index (Φ_{fairness}):** $\Delta P = P_{\text{retail}} - \text{RRP/D}$ and $\Phi_{\text{fairness}} = (\text{RRP/D}/P_{\text{retail}}) \times 100\%$.

10. **Step 10 — Multi-Bracket INDEX ($Y_{\text{DSP}} = \$618.85/\text{wk}$) & Confiscated NMW Labor ($w_{\text{NMW}} = \$26.44/\text{hr}$):** $\text{INDEX}_{\text{obs,DSP}} = (P_{\text{retail}}/618.85) \times 100\%$, $\text{INDEX}_{\text{RRP/D,DSP}} = (\text{RRP/D}/618.85) \times 100\%$, $\Delta\text{INDEX}_{\text{toll,DSP}} = (\Delta P/618.85) \times 100\%$, and $\Delta\tau_{\text{NMW}} = (\Delta P/26.44) \times 60$ min.

7.1.1 Physical & Operational Derivation of the Relativistic Urgency Parameter (v_{rel})

In the Universal Price Equation, $v_{\text{rel}} \in [0, 1)$ parameterizes the operational or biological urgency of an economic process, dictating the Lorentz coercion multiplier $\gamma(v_{\text{rel}}) = (1 - v_{\text{rel}}^2)^{-1/2}$. Rather than an arbitrary parameter, v_{rel} is derived from the ratio of operational slack tolerance to the critical deadline or temporal buffer:

$$v_{\text{rel}} = \frac{\tau_{\text{slack}}}{\tau_{\text{deadline}}} \in [0, 1) \tag{8}$$

- **Construction Critical Path ($v_{\text{rel}} = 0.16 \implies \gamma = 1.0131$):** Under AS 4100 and AS/NZS 3679.1, structural steel members (PFC channels, UB beams, UC columns, C-purlins) sit on the hard critical path of building erection. Crane hire ($300–$800/hr), rigger crews, and scaffolding cannot stand idle. Site laydown limits restrict on-site inventory buffering to ~ 1.6 days against a standard 10-day fabrication and delivery cycle: $v_{\text{rel}} \approx 1.6/10.0 = 0.16$, yielding $\gamma(0.16) = (1 - 0.0256)^{-1/2} = 1.0131$ (+1.31% urgency dilation on metabolic labor U).

- **Commercial SME Working Capital ($v_{\text{rel}} = 0.16 \implies \gamma = 1.0131$):** In Scale 4 bank lending, an operational farm or small enterprise maintains approximately 8 weeks of cash buffer against an annual 52-week operating cycle: $v_{\text{rel}} \approx 8/52 \approx 0.16$, enforcing an identical operational urgency without acute panic.

- **Bulk Non-Perishable Staples ($v_{\mathbf{rel}} = 0.05$–$0.06 \implies \gamma = 1.0013$–$1.0018$):** Dry pulse grains (faba beans, mungbeans, lentils) and solar sea salt carry 18–24+ month shelf buffers, giving near-zero purchasing urgency (+0.13% to +0.18%).

- **Chemical Setting & Structural Pours ($v_{\mathbf{rel}} = 0.20 \implies \gamma = 1.0206$):** Ready-mix concrete initial setting time (90 minutes) imposes an irreversible deadline where pour delays cause structural cold-joint failure (+2.06% dilation).

- **Acute Clinical Emergency ($v_{\mathbf{rel}} = 0.35$–$0.40 \implies \gamma = 1.0675$–$1.0911$):** Saline IV rehydration ($v_{\text{rel}} = 0.35$) and Salbutamol bronchospasm treatment ($v_{\text{rel}} = 0.40$) operate under acute life-or-death survival timelines, causing severe Lorentz labor dilation (+6.75% to +9.11%).

Representative Step-by-Step Worked Derivations (Physical Commodities)

- **Case 1: Commercial Sliced White Bread (680g Loaf):** Given $P_{\text{retail}} = \$4.10, P_e = \$0.38, P_b = \$0.45, U = 2.15, S = 1.00, R_n = 1.00, v_{\text{rel}} = 0.05$: (1) $P_{\text{phys}} = \$0.38 + \$0.45 = \$0.83$; (2) $\Delta_{\text{spread}} = \$4.10 - \$0.83 = \$3.27$; (3) $\gamma(0.05) = (1 - 0.0025)^{-1/2} = 1.0013$; (4) $W_{\text{num}} = 1.001252 \times 1.00 \times 2.15 = 2.1527$; (5) $M_{\text{pure}} = 2.1527/1.00 = \$2.1527$; (6) $M_{\text{fair}} = \$2.1527/0.98 = \$2.1966$; (7) RRP/D $= \$0.83 + \$2.1966 = \$\mathbf{3.03}$; (8) $R_a = 1 - (2.1527/3.27) = \mathbf{34.2}\%$; (9) $\Delta P = \$4.10 - \$3.03 = \$\mathbf{1.07}$, $\Phi_{\text{fairness}} = (3.0266/4.10) \times 100\% = \mathbf{73.8}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 0.66\%, \text{INDEX}_{\text{RRP/D,DSP}} = 0.49\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 0.17\%, \Delta\tau_{\text{NMW}} = (1.0734/26.44) \times 60 = \mathbf{2.44}$ min.

- **Case 47: Commercial Structural Steel Parallel Flange Channel (150x75mm PFC, per Metre):** Given $P_{\text{retail}} = \$65.50, P_e = \$14.50, P_b = \$19.80, U = 10.20, S = 1.10, R_n = 0.82, v_{\text{rel}} = 0.16$: (1) $P_{\text{phys}} = \$14.50 + \$19.80 = \$34.30$; (2) $\Delta_{\text{spread}} = \$65.50 - \$34.30 = \$31.20$; (3) $\gamma(0.16) = (1 - 0.0256)^{-1/2} = 1.0131$; (4) $W_{\text{num}} = 1.01305 \times 1.10 \times 10.20 = 11.3664$; (5) $M_{\text{pure}} = 11.3664/0.82 = \$13.8615$; (6) $M_{\text{fair}} = \$13.8615/0.98 = \$14.1444$; (7) RRP/D $= \$34.30 + \$14.1444 = \$\mathbf{48.44}$; (8) $R_a = 1 - (13.8615/31.20) = \mathbf{55.6}\%$; (9) $\Delta P = \$65.50 - \$48.44 = \$\mathbf{17.06}$, $\Phi_{\text{fairness}} = (48.4444/65.50) \times 100\% = \mathbf{74.0}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 10.58\%, \text{INDEX}_{\text{RRP/D,DSP}} = 7.83\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 2.76\%, \Delta\tau_{\text{NMW}} = (17.0556/26.44) \times 60 = \mathbf{38.70}$ min.

- **Case 48: Australian Narrow-Leaf Lupins (Bulk Feed Lupins, per Tonne):** Given $P_{\text{retail}} = \$520.00, P_e = \$131.00, P_b = \$121.00, U = 112.00, S = 1.10, R_n = 1.08, v_{\text{rel}} = 0.05$: (1) $P_{\text{phys}} = \$131.00 + \$121.00 = \$252.00$; (2) $\Delta_{\text{spread}} = \$520.00 - \$252.00 = \$268.00$; (3) $\gamma(0.05) = (1 - 0.0025)^{-1/2} = 1.0013$; (4) $W_{\text{num}} = 1.001252 \times 1.10 \times 112.00 = 123.3543$; (5) $M_{\text{pure}} = 123.3543/1.08 = \$114.2169$; (6) $M_{\text{fair}} = \$114.2169/0.98 = \$116.5479$; (7) RRP/D $= \$252.00 + \$116.5479 = \$\mathbf{368.55}$; (8) $R_a = 1 - (114.2169/268.00) = \mathbf{57.4}\%$; (9) $\Delta P = \$520.00 - \$368.55 = \$\mathbf{151.45}$, $\Phi_{\text{fairness}} = (368.5479/520.00) \times 100\% = \mathbf{70.9}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 84.03\%, \text{INDEX}_{\text{RRP/D,DSP}} = 59.55\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 24.47\%, \Delta\tau_{\text{NMW}} = (151.4521/26.44) \times 60 = \mathbf{343.69}$ min (5.73 hr).

- **Case 49: Commercial Galvanized Steel Built-In Lintel / T-Bar (200x10mm Lintel, per Metre):** Given $P_{\text{retail}} = \$105.00, P_e = \$22.80, P_b = \$33.60, U = 16.50, S = 1.10, R_n = 0.82, v_{\text{rel}} = 0.18$: (1) $P_{\text{phys}} = \$22.80 + \$33.60 = \$56.40$; (2) $\Delta_{\text{spread}} = \$105.00 - \$56.40 = \$48.60$; (3) $\gamma(0.18) = (1 - 0.0324)^{-1/2} = 1.0166$; (4) $W_{\text{num}} = 1.0166 \times 1.10 \times 16.50 = 18.4514$; (5) $M_{\text{pure}} = 18.4514/0.82 = \$22.5017$; (6)

$M_{\text{fair}} = \$22.5017/0.98 = \$22.9609$; (7) RRP/D $= \$56.40 + \$22.9609 = \$\mathbf{79.36}$; (8) $R_a = 1-(22.5017/48.60) = \mathbf{53.7}\%$; (9) $\Delta P = \$105.00-\$79.36 = \$\mathbf{25.64}$, $\Phi_{\text{fairness}} = (79.3609/105.00)\times 100\% = \mathbf{75.6}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 16.97\%$, $\text{INDEX}_{\text{RRP/D,DSP}} = 12.82\%$, $\Delta\text{INDEX}_{\text{toll,DSP}} = 4.14\%$, $\Delta\tau_{\text{NMW}} = (25.6391/26.44) \times 60 = \mathbf{58.18}$ min.

- **Case 50: Australian Mungbeans (Berken / Crystal Type Pulse Grain, per Tonne):** Given $P_{\text{retail}} = \$1150.00$, $P_e = \$175.00$, $P_b = \$185.00$, $U = 165.00$, $S = 1.15$, $R_n = 1.04$, $v_{\text{rel}} = 0.08$: (1) $P_{\text{phys}} = \$175.00 + \$185.00 = \$360.00$; (2) $\Delta_{\text{spread}} = \$1150.00 - \$360.00 = \$790.00$; (3) $\gamma(0.08) = (1 - 0.0064)^{-1/2} = 1.0032$; (4) $W_{\text{num}} = 1.0032 \times 1.15 \times 165.00 = 190.3601$; (5) $M_{\text{pure}} = 190.3601/1.04 = \$183.0386$; (6) $M_{\text{fair}} = \$183.0386/0.98 = \$186.7741$; (7) RRP/D $= \$360.00+\$186.7741 = \$\mathbf{546.77}$; (8) $R_a = 1 - (183.0386/790.00) = \mathbf{76.8}\%$; (9) $\Delta P = \$1150.00 - \$546.77 = \$\mathbf{603.23}$, $\Phi_{\text{fairness}} = (546.7741/1150.00) \times 100\% = \mathbf{47.5}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 185.83\%$, $\text{INDEX}_{\text{RRP/D,DSP}} = 88.35\%$, $\Delta\text{INDEX}_{\text{toll,DSP}} = 97.48\%$, $\Delta\tau_{\text{NMW}} = (603.2259/26.44)\times 60 = \mathbf{1368.89}$ min (22.81 hr).

- **Case 51: Commercial Structural Steel Parallel Flange Channel (200x75mm PFC, per Metre):** Given $P_{\text{retail}} = \$85.00$, $P_e = \$18.80$, $P_b = \$25.60$, $U = 13.20$, $S = 1.10$, $R_n = 0.82$, $v_{\text{rel}} = 0.16$: (1) $P_{\text{phys}} = \$18.80 + \$25.60 = \$44.40$; (2) $\Delta_{\text{spread}} = \$85.00 - \$44.40 = \$40.60$; (3) $\gamma(0.16) = (1 - 0.0256)^{-1/2} = 1.0131$; (4) $W_{\text{num}} = 1.01305 \times 1.10 \times 13.20 = 14.7095$; (5) $M_{\text{pure}} = 14.7095/0.82 = \$17.9384$; (6) $M_{\text{fair}} = \$17.9384/0.98 = \$18.3045$; (7) RRP/D $= \$44.40 + \$18.3045 = \$\mathbf{62.70}$; (8) $R_a = 1-(17.9384/40.60) = \mathbf{55.8}\%$; (9) $\Delta P = \$85.00-\$62.70 = \$\mathbf{22.30}$, $\Phi_{\text{fairness}} = (62.7045/85.00)\times 100\% = \mathbf{73.8}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 13.74\%$, $\text{INDEX}_{\text{RRP/D,DSP}} = 10.13\%$, $\Delta\text{INDEX}_{\text{toll,DSP}} = 3.60\%$, $\Delta\tau_{\text{NMW}} = (22.2955/26.44) \times 60 = \mathbf{50.59}$ min.

- **Case 52: Commercial De-Hulled Split Faba Beans (Machine Dressed Export & Food Grade, per Tonne):** Given $P_{\text{retail}} = \$750.00$, $P_e = \$226.00$, $P_b = \$156.00$, $U = 138.00$, $S = 1.12$, $R_n = 1.05$, $v_{\text{rel}} = 0.06$: (1) $P_{\text{phys}} = \$226.00 + \$156.00 = \$382.00$; (2) $\Delta_{\text{spread}} = \$750.00 - \$382.00 = \$368.00$; (3) $\gamma(0.06) = (1-0.0036)^{-1/2} = 1.0018$; (4) $W_{\text{num}} = 1.0018\times 1.12\times 138.00 = 154.8390$; (5) $M_{\text{pure}} = 154.8390/1.05 = \$147.4657$; (6) $M_{\text{fair}} = \$147.4657/0.98 = \$150.4752$; (7) RRP/D $= \$382.00+\$150.4752 = \$\mathbf{532.48}$; (8) $R_a = 1-(147.4657/368.00) = \mathbf{59.9}\%$; (9) $\Delta P = \$750.00 - \$532.48 = \$\mathbf{217.52}$, $\Phi_{\text{fairness}} = (532.4752/750.00) \times 100\% = \mathbf{71.0}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 121.19\%$, $\text{INDEX}_{\text{RRP/D,DSP}} = 86.04\%$, $\Delta\text{INDEX}_{\text{toll,DSP}} = 35.15\%$, $\Delta\tau_{\text{NMW}} = (217.5248/26.44) \times 60 = \mathbf{493.63}$ min (8.23 hr).

- **Case 53: Commercial Structural Steel Parallel Flange Channel (250x90mm PFC, per Metre):** Given $P_{\text{retail}} = \$132.00$, $P_e = \$29.20$, $P_b = \$39.70$, $U = 20.40$, $S = 1.10$, $R_n = 0.82$, $v_{\text{rel}} = 0.16$: (1) $P_{\text{phys}} = \$29.20 + \$39.70 = \$68.90$; (2) $\Delta_{\text{spread}} = \$132.00 - \$68.90 = \$63.10$; (3) $\gamma(0.16) = (1 - 0.0256)^{-1/2} = 1.0131$; (4) $W_{\text{num}} = 1.01305\times 1.10\times 20.40 = 22.7329$; (5) $M_{\text{pure}} = 22.7329/0.82 = \$27.7230$; (6) $M_{\text{fair}} = \$27.7230/0.98 = \$28.2888$; (7) RRP/D $= \$68.90 + \$28.2888 = \$\mathbf{97.19}$; (8) $R_a = 1-(27.7230/63.10) = \mathbf{56.1}\%$; (9) $\Delta P = \$132.00-\$97.19 = \$\mathbf{34.81}$, $\Phi_{\text{fairness}} = (97.1888/132.00)\times 100\% = \mathbf{73.6}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 21.33\%$, $\text{INDEX}_{\text{RRP/D,DSP}} = 15.70\%$, $\Delta\text{INDEX}_{\text{toll,DSP}} = 5.63\%$, $\Delta\tau_{\text{NMW}} = (34.8112/26.44)\times 60 = \mathbf{79.00}$ min (1.32 hr).

- **Case 54: Commercial De-Hulled Split Yellow Field Peas (Machine Dressed Food Grade, per Tonne):** Given $P_{\text{retail}} = \$680.00$, $P_e = \$207.00$, $P_b = \$145.00$, $U = 132.00$, $S = 1.10$, $R_n = 1.05$, $v_{\text{rel}} = 0.05$: (1) $P_{\text{phys}} = \$207.00 + \$145.00 = \$352.00$;

(2) $\Delta_{\text{spread}} = \$680.00 - \$352.00 = \$328.00$; (3) $\gamma(0.05) = (1 - 0.0025)^{-1/2} = 1.0013$; (4) $W_{\text{num}} = 1.00125 \times 1.10 \times 132.00 = 145.3818$; (5) $M_{\text{pure}} = 145.3818/1.05 = \$138.4589$; (6) $M_{\text{fair}} = \$138.4589/0.98 = \$141.2846$; (7) RRP/D $= \$352.00 + \$141.2846 = \$\mathbf{493.28}$; (8) $R_a = 1 - (138.4589/328.00) = \mathbf{57.8}\%$; (9) $\Delta P = \$680.00 - \$493.28 = \$\mathbf{186.72}$, $\Phi_{\text{fairness}} = (493.2846/680.00) \times 100\% = \mathbf{72.5}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 109.88\%, \text{INDEX}_{\text{RRP/D,DSP}} = 79.71\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 30.17\%, \Delta\tau_{\text{NMW}} = (186.7154/26.44) \times 60 = \mathbf{423.71}$ min (7.06 hr).

- **Case 55: Commercial Structural Steel Parallel Flange Channel (300x90mm PFC, per Metre):** Given $P_{\text{retail}} = \$149.50, P_e = \$33.20, P_b = \$45.20, U = 23.10, S = 1.10, R_n = 0.82, v_{\text{rel}} = 0.16$: (1) $P_{\text{phys}} = \$33.20 + \$45.20 = \$78.40$; (2) $\Delta_{\text{spread}} = \$149.50 - \$78.40 = \$71.10$; (3) $\gamma(0.16) = (1 - 0.0256)^{-1/2} = 1.0131$; (4) $W_{\text{num}} = 1.0131 \times 1.10 \times 23.10 = 25.7416$; (5) $M_{\text{pure}} = 25.7416/0.82 = \$31.3922$; (6) $M_{\text{fair}} = \$31.3922/0.98 = \$32.0329$; (7) RRP/D $= \$78.40 + \$32.0329 = \$\mathbf{110.43}$; (8) $R_a = 1 - (31.3922/71.10) = \mathbf{55.8}\%$; (9) $\Delta P = \$149.50 - \$110.43 = \$\mathbf{39.07}$, $\Phi_{\text{fairness}} = (110.4329/149.50) \times 100\% = \mathbf{73.9}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 24.16\%, \text{INDEX}_{\text{RRP/D,DSP}} = 17.84\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 6.31\%, \Delta\tau_{\text{NMW}} = (39.0671/26.44) \times 60 = \mathbf{88.65}$ min.

- **Case 56: Commercial De-Hulled Split Red Lentils (Machine Dressed Export Grade, per Tonne):** Given $P_{\text{retail}} = \$1050.00, P_e = \$262.00, P_b = \$178.00, U = 146.00, S = 1.15, R_n = 1.05, v_{\text{rel}} = 0.07$: (1) $P_{\text{phys}} = \$262.00 + \$178.00 = \$440.00$; (2) $\Delta_{\text{spread}} = \$1050.00 - \$440.00 = \$610.00$; (3) $\gamma(0.07) = (1 - 0.0049)^{-1/2} = 1.0025$; (4) $W_{\text{num}} = 1.0025 \times 1.15 \times 146.00 = 168.3129$; (5) $M_{\text{pure}} = 168.3129/1.05 = \$160.2980$; (6) $M_{\text{fair}} = \$160.2980/0.98 = \$163.5694$; (7) RRP/D $= \$440.00 + \$163.5694 = \$\mathbf{603.57}$; (8) $R_a = 1 - (160.2980/610.00) = \mathbf{73.7}\%$; (9) $\Delta P = \$1050.00 - \$603.57 = \$\mathbf{446.43}$, $\Phi_{\text{fairness}} = (603.5694/1050.00) \times 100\% = \mathbf{57.5}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 169.67\%, \text{INDEX}_{\text{RRP/D,DSP}} = 97.53\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 72.14\%, \Delta\tau_{\text{NMW}} = (446.4306/26.44) \times 60 = \mathbf{1013.08}$ min (16.88 hr).

- **Case 57: Pharmaceutical Saline IV Infusion Bag (Sodium Chloride 0.9%, 1,000 mL Bag):** Given $P_{\text{retail}} = \$8.50, P_e = \$0.65, P_b = \$0.85, U = 2.40, S = 1.20, R_n = 0.90, v_{\text{rel}} = 0.35$: (1) $P_{\text{phys}} = \$0.65 + \$0.85 = \$1.50$; (2) $\Delta_{\text{spread}} = \$8.50 - \$1.50 = \$7.00$; (3) $\gamma(0.35) = (1 - 0.1225)^{-1/2} = 1.0675$; (4) $W_{\text{num}} = 1.0675 \times 1.20 \times 2.40 = 3.0745$; (5) $M_{\text{pure}} = 3.0745/0.90 = \$3.4161$; (6) $M_{\text{fair}} = \$3.4161/0.98 = \$3.4858$; (7) RRP/D $= \$1.50 + \$3.4858 = \$\mathbf{4.99}$; (8) $R_a = 1 - (3.4161/7.00) = \mathbf{51.2}\%$; (9) $\Delta P = \$8.50 - \$4.99 = \$\mathbf{3.51}$, $\Phi_{\text{fairness}} = (4.9858/8.50) \times 100\% = \mathbf{58.7}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 1.37\%, \text{INDEX}_{\text{RRP/D,DSP}} = 0.81\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 0.57\%, \Delta\tau_{\text{NMW}} = (3.5142/26.44) \times 60 = \mathbf{7.97}$ min.

- **Case 58: Commercial Solar PV Module (440W Monocrystalline Residential Panel, Tier-1):** Given $P_{\text{retail}} = \$165.00, P_e = \$38.50, P_b = \$52.00, U = 14.50, S = 1.08, R_n = 0.88, v_{\text{rel}} = 0.12$: (1) $P_{\text{phys}} = \$38.50 + \$52.00 = \$90.50$; (2) $\Delta_{\text{spread}} = \$165.00 - \$90.50 = \$74.50$; (3) $\gamma(0.12) = (1 - 0.0144)^{-1/2} = 1.0073$; (4) $W_{\text{num}} = 1.0073 \times 1.08 \times 14.50 = 15.7740$; (5) $M_{\text{pure}} = 15.7740/0.88 = \$17.9250$; (6) $M_{\text{fair}} = \$17.9250/0.98 = \$18.2908$; (7) RRP/D $= \$90.50 + \$18.2908 = \$\mathbf{108.79}$; (8) $R_a = 1 - (17.9250/74.50) = \mathbf{75.9}\%$; (9) $\Delta P = \$165.00 - \$108.79 = \$\mathbf{56.21}$, $\Phi_{\text{fairness}} = (108.7908/165.00) \times 100\% = \mathbf{65.9}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 26.66\%, \text{INDEX}_{\text{RRP/D,DSP}} = 17.58\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 9.08\%, \Delta\tau_{\text{NMW}} = (56.2092/26.44) \times 60 = \mathbf{127.55}$ min (2.13 hr).

- **Case 59: Commercial Structural Steel Universal Column (200UC46.2, AS/NZS 3679.1 Grade 300PLUS):** Given $P_{\text{retail}} = \$172.00, P_e = \$38.50, P_b = \$52.00, U = 26.50, S = 1.10, R_n = 0.82, v_{\text{rel}} = 0.16$: (1) $P_{\text{phys}} = \$38.50 + \$52.00 = \$90.50$; (2) $\Delta_{\text{spread}} = \$172.00 - \$90.50 = \$81.50$; (3) $\gamma(0.16) = (1 - 0.0256)^{-1/2} = 1.0131$; (4) $W_{\text{num}} = 1.013051 \times 1.10 \times 26.50 = 29.5304$; (5) $M_{\text{pure}} = 29.5304/0.82 = \$36.0127$; (6) $M_{\text{fair}} = \$36.0127/0.98 = \$36.7477$; (7) RRP/D $= \$90.50 + \$36.7477 = \$\mathbf{127.25}$; (8) $R_a = 1 - (36.0127/81.50) = \mathbf{55.8}\%$; (9) $\Delta P = \$172.00 - \$127.25 = \$\mathbf{44.75}$, $\Phi_{\text{fairness}} = (127.2477/172.00) \times 100\% = \mathbf{74.0}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 27.79\%, \text{INDEX}_{\text{RRP/D,DSP}} = 20.56\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 7.23\%, \Delta\tau_{\text{NMW}} = (44.7523/26.44) \times 60 = \mathbf{101.56}$ min (1.69 hr).

- **Case 60: Commercial Australian Faba Bean Protein Concentrate / Flour (min 55% Protein, per Tonne):** Given $P_{\text{retail}} = \$1280.00, P_e = \$310.00, P_b = \$245.00, U = 162.00, S = 1.15, R_n = 1.05, v_{\text{rel}} = 0.06$: (1) $P_{\text{phys}} = \$310.00 + \$245.00 = \$555.00$; (2) $\Delta_{\text{spread}} = \$1280.00 - \$555.00 = \$725.00$; (3) $\gamma(0.06) = (1 - 0.0036)^{-1/2} = 1.0018$; (4) $W_{\text{num}} = 1.001805 \times 1.15 \times 162.00 = 186.6362$; (5) $M_{\text{pure}} = 186.6362/1.05 = \$177.7488$; (6) $M_{\text{fair}} = \$177.7488/0.98 = \$181.3763$; (7) RRP/D $= \$555.00 + \$181.3763 = \$\mathbf{736.38}$; (8) $R_a = 1 - (177.7488/725.00) = \mathbf{75.5}\%$; (9) $\Delta P = \$1280.00 - \$736.38 = \$\mathbf{543.62}$, $\Phi_{\text{fairness}} = (736.3763/1280.00) \times 100\% = \mathbf{57.5}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 206.84\%, \text{INDEX}_{\text{RRP/D,DSP}} = 118.99\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 87.84\%, \Delta\tau_{\text{NMW}} = (543.6237/26.44) \times 60 = \mathbf{1233.64}$ min (20.56 hr).

- **Case 61: Commercial Structural Steel Universal Column (250UC72.9, AS/NZS 3679.1 Grade 300PLUS):** Given $P_{\text{retail}} = \$272.00, P_e = \$60.50, P_b = \$82.50, U = 41.50, S = 1.10, R_n = 0.82, v_{\text{rel}} = 0.16$: (1) $P_{\text{phys}} = \$60.50 + \$82.50 = \$143.00$; (2) $\Delta_{\text{spread}} = \$272.00 - \$143.00 = \$129.00$; (3) $\gamma(0.16) = (1 - 0.0256)^{-1/2} = 1.0131$; (4) $W_{\text{num}} = 1.013051 \times 1.10 \times 41.50 = 46.2458$; (5) $M_{\text{pure}} = 46.2458/0.82 = \$56.3973$; (6) $M_{\text{fair}} = \$56.3973/0.98 = \$57.5483$; (7) RRP/D $= \$143.00 + \$57.5483 = \$\mathbf{200.55}$; (8) $R_a = 1 - (56.3973/129.00) = \mathbf{56.3}\%$; (9) $\Delta P = \$272.00 - \$200.55 = \$\mathbf{71.45}$, $\Phi_{\text{fairness}} = (200.5483/272.00) \times 100\% = \mathbf{73.7}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 43.95\%, \text{INDEX}_{\text{RRP/D,DSP}} = 32.41\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 11.55\%, \Delta\tau_{\text{NMW}} = (71.4517/26.44) \times 60 = \mathbf{162.14}$ min (2.70 hr).

- **Case 62: Commercial Australian Field Pea Protein Isolate (min 80% Protein, Spray-Dried, per Tonne):** Given $P_{\text{retail}} = \$3450.00, P_e = \$820.00, P_b = \$615.00, U = 385.00, S = 1.15, R_n = 1.05, v_{\text{rel}} = 0.06$: (1) $P_{\text{phys}} = \$820.00 + \$615.00 = \$1435.00$; (2) $\Delta_{\text{spread}} = \$3450.00 - \$1435.00 = \$2015.00$; (3) $\gamma(0.06) = (1 - 0.0036)^{-1/2} = 1.0018$; (4) $W_{\text{num}} = 1.001805 \times 1.15 \times 385.00 = 443.5491$; (5) $M_{\text{pure}} = 443.5491/1.05 = \$422.4277$; (6) $M_{\text{fair}} = \$422.4277/0.98 = \$431.0487$; (7) RRP/D $= \$1435.00 + \$431.0487 = \$\mathbf{1866.05}$; (8) $R_a = 1 - (422.4277/2015.00) = \mathbf{79.0}\%$; (9) $\Delta P = \$3450.00 - \$1866.05 = \$\mathbf{1583.95}$, $\Phi_{\text{fairness}} = (1866.0487/3450.00) \times 100\% = \mathbf{54.1}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 557.49\%, \text{INDEX}_{\text{RRP/D,DSP}} = 301.54\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 255.95\%, \Delta\tau_{\text{NMW}} = (1583.9513/26.44) \times 60 = \mathbf{3594.44}$ min (59.91 hr).

- **Case 63: Commercial Structural Steel Universal Column (310UC96.8, AS/NZS 3679.1 Grade 300PLUS):** Given $P_{\text{retail}} = \$360.00, P_e = \$80.50, P_b = \$109.50, U = 55.00, S = 1.10, R_n = 0.82, v_{\text{rel}} = 0.16$: (1) $P_{\text{phys}} = \$80.50 + \$109.50 = \$190.00$; (2) $\Delta_{\text{spread}} = \$360.00 - \$190.00 = \$170.00$; (3) $\gamma(0.16) = (1 - 0.0256)^{-1/2} = 1.0131$; (4) $W_{\text{num}} = 1.013051 \times 1.10 \times 55.00 = 61.2896$; (5) $M_{\text{pure}} = 61.2896/0.82 = \$74.7434$; (6) $M_{\text{fair}} = \$74.7434/0.98 = \$76.2688$; (7) RRP/D $= \$190.00 + \$76.2688 =$

$\mathbf{266.27}$; (8) $R_a = 1 - (74.7434/170.00) = \mathbf{56.0}\%$; (9) $\Delta P = \$360.00 - \$266.27 = \mathbf{\$93.73}$, $\Phi_{\text{fairness}} = (266.2688/360.00) \times 100\% = \mathbf{74.0}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 58.17\%, \text{INDEX}_{\text{RRP/D,DSP}} = 43.03\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 15.15\%, \Delta\tau_{\text{NMW}} = (93.7312/26.44) \times 60 = \mathbf{212.70}$ min (3.55 hr).

- **Case 64: Commercial Australian Chickpea Protein Isolate / Functional Flour (min 75% Protein, Alkaline-Extracted, per Tonne):** Given $P_{\text{retail}} = \$3150.00, P_e = \$745.00, P_b = \$560.00, U = 350.00, S = 1.15, R_n = 1.04, v_{\text{rel}} = 0.06$: (1) $P_{\text{phys}} = \$745.00 + \$560.00 = \$1305.00$; (2) $\Delta_{\text{spread}} = \$3150.00 - \$1305.00 = \$1845.00$; (3) $\gamma(0.06) = (1 - 0.0036)^{-1/2} = 1.0018$; (4) $W_{\text{num}} = 1.001805 \times 1.15 \times 350.00 = 403.2265$; (5) $M_{\text{pure}} = 403.2265/1.04 = \$387.7178$; (6) $M_{\text{fair}} = \$387.7178/0.98 = \$395.6304$; (7) RRP/D = \$1305.00+\$395.6304 = $\mathbf{\$1700.63}$; (8) $R_a = 1-(387.7178/1845.00) = \mathbf{79.0}\%$; (9) $\Delta P = \$3150.00-\$1700.63 = \mathbf{\$1449.37}$, $\Phi_{\text{fairness}} = (1700.6304/3150.00) \times 100\% = \mathbf{54.0}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 509.01\%, \text{INDEX}_{\text{RRP/D,DSP}} = 274.80\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 234.20\%, \Delta\tau_{\text{NMW}} = (1449.3696/26.44) \times 60 = \mathbf{3289.04}$ min (54.82 hr).

- **Case 65: Commercial Structural Steel Universal Column (310UC118, AS/NZS 3679.1 Grade 300PLUS):** Given $P_{\text{retail}} = \$438.00, P_e = \$98.00, P_b = \$133.50, U = 67.00, S = 1.10, R_n = 0.82, v_{\text{rel}} = 0.16$: (1) $P_{\text{phys}} = \$98.00 + \$133.50 = \$231.50$; (2) $\Delta_{\text{spread}} = \$438.00 - \$231.50 = \$206.50$; (3) $\gamma(0.16) = (1-0.0256)^{-1/2} = 1.0131$; (4) $W_{\text{num}} = 1.013051 \times 1.10 \times 67.00 = 74.6619$; (5) $M_{\text{pure}} = 74.6619/0.82 = \$91.0511$; (6) $M_{\text{fair}} = \$91.0511/0.98 = \$92.9092$; (7) RRP/D = \$231.50+\$92.9092 = $\mathbf{\$324.41}$; (8) $R_a = 1 - (91.0511/206.50) = \mathbf{55.9}\%$; (9) $\Delta P = \$438.00 - \$324.41 = \mathbf{\$113.59}$, $\Phi_{\text{fairness}} = (324.4092/438.00) \times 100\% = \mathbf{74.1}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 70.78\%, \text{INDEX}_{\text{RRP/D,DSP}} = 52.42\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 18.36\%, \Delta\tau_{\text{NMW}} = (113.5908/26.44) \times 60 = \mathbf{257.77}$ min (4.30 hr).

- **Case 66: Commercial Australian Faba Bean Protein Isolate (min 85% Protein, Wet-Extracted & Spray-Dried, per Tonne):** Given $P_{\text{retail}} = \$3580.00, P_e = \$855.00, P_b = \$640.00, U = 398.00, S = 1.15, R_n = 1.05, v_{\text{rel}} = 0.06$: (1) $P_{\text{phys}} = \$855.00 + \$640.00 = \$1495.00$; (2) $\Delta_{\text{spread}} = \$3580.00 - \$1495.00 = \$2085.00$; (3) $\gamma(0.06) = (1 - 0.0036)^{-1/2} = 1.0018$; (4) $W_{\text{num}} = 1.001805 \times 1.15 \times 398.00 = 458.5261$; (5) $M_{\text{pure}} = 458.5261/1.05 = \$436.6915$; (6) $M_{\text{fair}} = \$436.6915/0.98 = \$445.6036$; (7) RRP/D = \$1495.00+\$445.6036 = $\mathbf{\$1940.60}$; (8) $R_a = 1-(436.6915/2085.00) = \mathbf{79.1}\%$; (9) $\Delta P = \$3580.00-\$1940.60 = \mathbf{\$1639.40}$, $\Phi_{\text{fairness}} = (1940.6036/3580.00) \times 100\% = \mathbf{54.2}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 578.49\%, \text{INDEX}_{\text{RRP/D,DSP}} = 313.58\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 264.91\%, \Delta\tau_{\text{NMW}} = (1639.3964/26.44) \times 60 = \mathbf{3720.26}$ min (62.00 hr).

- **Case 67: Commercial Structural Steel Universal Column (310UC158, AS/NZS 3679.1 Grade 300PLUS):** Given $P_{\text{retail}} = \$588.00, P_e = \$131.50, P_b = \$179.50, U = 90.00, S = 1.10, R_n = 0.82, v_{\text{rel}} = 0.16$: (1) $P_{\text{phys}} = \$131.50 + \$179.50 = \$311.00$; (2) $\Delta_{\text{spread}} = \$588.00 - \$311.00 = \$277.00$; (3) $\gamma(0.16) = (1 - 0.0256)^{-1/2} = 1.0131$; (4) $W_{\text{num}} = 1.013051 \times 1.10 \times 90.00 = 100.2921$; (5) $M_{\text{pure}} = 100.2921/0.82 = \$122.3074$; (6) $M_{\text{fair}} = \$122.3074/0.98 = \$124.8035$; (7) RRP/D = \$311.00+\$124.8035 = $\mathbf{\$435.80}$; (8) $R_a = 1-(122.3074/277.00) = \mathbf{55.8}\%$; (9) $\Delta P = \$588.00 - \$435.80 = \mathbf{\$152.20}$, $\Phi_{\text{fairness}} = (435.8035/588.00) \times 100\% = \mathbf{74.1}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 95.01\%, \text{INDEX}_{\text{RRP/D,DSP}} = 70.42\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 24.59\%, \Delta\tau_{\text{NMW}} = (152.1965/26.44) \times 60 = \mathbf{345.38}$ min (5.76 hr).

- **Case 68: Commercial Australian Lupin Protein Isolate (min 88% Protein, Aqueous-Extracted & Spray-Dried, per Tonne):** Given $P_{\text{retail}} = \$3750.00, P_e = \$895.00, P_b = \$670.00, U = 415.00, S = 1.15, R_n = 1.08, v_{\text{rel}} = 0.06$: (1) $P_{\text{phys}} = \$895.00 + \$670.00 = \$1565.00$; (2) $\Delta_{\text{spread}} = \$3750.00 - \$1565.00 = \$2185.00$; (3) $\gamma(0.06) = (1 - 0.0036)^{-1/2} = 1.0018$; (4) $W_{\text{num}} = 1.001805 \times 1.15 \times 415.00 = 478.1114$; (5) $M_{\text{pure}} = 478.1114/1.08 = \$442.6957$; (6) $M_{\text{fair}} = \$442.6957/0.98 = \$451.7303$; (7) RRP/D = \$1565.00+\$451.7303 = \$**2016.73**; (8) $R_a = 1-(442.6957/2185.00) =$ **79.7**%; (9) $\Delta P = \$3750.00-\$2016.73 = \$$**1733.27**, $\Phi_{\text{fairness}} = (2016.7303/3750.00)\times 100\% =$ **53.8**%; (10) $\text{INDEX}_{\text{obs,DSP}} = 605.96\%, \text{INDEX}_{\text{RRP/D,DSP}} = 325.88\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 280.08\%, \Delta\tau_{\text{NMW}} = (1733.2697/26.44) \times 60 =$ **3933.29** min (65.55 hr).

- **Case 69: Commercial Structural Steel Universal Beam (200UB18.2, AS/NZS 3679.1 Grade 300PLUS):** Given $P_{\text{retail}} = \$68.50, P_e = \$15.20, P_b = \$20.80, U = 10.60, S = 1.11, R_n = 0.82, v_{\text{rel}} = 0.16$: (1) $P_{\text{phys}} = \$15.20 + \$20.80 = \$36.00$; (2) $\Delta_{\text{spread}} = \$68.50 - \$36.00 = \$32.50$; (3) $\gamma(0.16) = \frac{1}{\sqrt{1-0.16^2}} = 1.0131$; (4) $W_{\text{num}} = 1.0131 \times 1.11 \times 10.60 = 11.9196$; (5) $M_{\text{pure}} = 11.9196/0.82 = \$14.5360$; (6) $M_{\text{fair}} = \$14.5360/0.98 = \$14.8327$; (7) RRP/D = \$36.00 + \$14.8327 = \$**50.83**; (8) $R_a = 1-(14.5360/32.50) =$ **55.3**%; (9) $\Delta P = \$68.50-\$50.83 = \$$**17.67**, $\Phi_{\text{fairness}} = (50.8327/68.50)\times 100\% =$ **74.2**%; (10) $\text{INDEX}_{\text{obs,DSP}} = 11.07\%, \text{INDEX}_{\text{RRP/D,DSP}} = 8.21\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 2.85\%, \Delta\tau_{\text{NMW}} = (17.6673/26.44)\times 60 =$ **40.09** min (0.67 hr).

- **Case 70: Commercial Australian Oat Milk Base / Hydrolyzed Oat Flour (min 15% Beta-Glucan / Soluble Fiber, per Tonne):** Given $P_{\text{retail}} = \$2450.00, P_e = \$580.00, P_b = \$440.00, U = 285.00, S = 1.14, R_n = 1.04, v_{\text{rel}} = 0.06$: (1) $P_{\text{phys}} = \$580.00 + \$440.00 = \$1020.00$; (2) $\Delta_{\text{spread}} = \$2450.00 - \$1020.00 = \$1430.00$; (3) $\gamma(0.06) = \frac{1}{\sqrt{1-0.06^2}} = 1.0018$; (4) $W_{\text{num}} = 1.0018 \times 1.14 \times 285.00 = 325.4864$; (5) $M_{\text{pure}} = 325.4864/1.04 = \$312.9677$; (6) $M_{\text{fair}} = \$312.9677/0.98 = \$319.3548$; (7) RRP/D = \$1020.00 + \$319.3548 = \$**1339.35**; (8) $R_a = 1 - (312.9677/1430.00) =$ **78.1**%; (9) $\Delta P = \$2450.00-\$1339.35 = \$$**1110.65**, $\Phi_{\text{fairness}} = (1339.3548/2450.00)\times 100\% =$ **54.7**%; (10) $\text{INDEX}_{\text{obs,DSP}} = 395.90\%, \text{INDEX}_{\text{RRP/D,DSP}} = 216.43\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 179.47\%, \Delta\tau_{\text{NMW}} = (1110.6452/26.44) \times 60 =$ **2520.37** min (42.01 hr).

7.2 Complete 11-Step Mathematical Working Out for All 5 Big 4 Bank Loan Scales

For commercial bank credit facilities (where principal is L, nominal interest rate is r_{nom}, monthly account/service fee is F_{mo}, and the RBA Cash Rate is 4.60% p.a.), Step 11 normalizes monthly charges into effective annual percentage rates and costs per $\$10,000$ borrowed:

- **Scale 1 — Individual Revolving Credit Card (\$5,000 Balance @ 20.99% p.a. + \$3.00/mo Fee):** Monthly Interest = $(\$5,000 \times 0.2099)/12 = \$87.46$; $P_{\text{retail}} = \$87.46 + \$3.00 = \$90.46$/mo ($\$1,085.52$/yr). (1) $P_{\text{phys}} = \$1.20 + \$8.20 = \$9.40$ (where $\$7.50 = 1.80\%$ p.a. mutual default pool on \$5k); (2) $\Delta_{\text{spread}} = \$90.46-\$9.40 = \$81.06$; (3) $\gamma(0.32) = (1 - 0.32^2)^{-1/2} = 1.0555$; (4) $W_{\text{num}} = 1.0555 \times 1.02 \times 3.50 = 3.7681$; (5) $M_{\text{pure}} = 3.7681/0.98 = \$3.8450$; (6) $M_{\text{fair}} = \$3.8450/0.98 = \$3.9235$; (7) RRP/D = $\$9.40 + \$3.9235 =$ \$**13.32**/mo ($\$159.84$/yr); (8) $R_a = 1-(3.8450/81.06) =$ **95.3**%; (9) $\Delta P = \$90.46-\$13.32 = \$$**77.14**/mo ($\$925.68$/yr),

$\Phi_{\text{fairness}} = \mathbf{14.7}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 14.62\%, \text{INDEX}_{\text{RRP/D,DSP}} = 2.15\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 12.46\%, \Delta\tau_{\text{NMW}} = \mathbf{175.04}$ min; (11) Effective Rate = $(\$1,085.52/\$5,000) \times 100\% = \mathbf{21.71}\%$ p.a.; Fair RRP/D Rate = $(\$159.84/\$5,000) \times 100\% = \mathbf{3.20}\%$ p.a.; RBA Spread = $(20.99 - 4.60) \times 100 = +\mathbf{1,639}$ bps; Per \$10k Borrowed ($\times$2.0): Obs = $\$\mathbf{2,171.00}$/yr, Fair = $\$\mathbf{319.68}$/yr, Extractive Toll = $\$\mathbf{1,851.32}$/yr.

- **Scale 2 — Individual Unsecured Personal Loan (\$20,000 Facility @ 14.50% p.a. + \$12.00/mo Fee):** Monthly Interest = $(\$20,000 \times 0.1450)/12 = \$241.67$; $P_{\text{retail}} = \$241.67 + \$12.00 = \$253.67$/mo ($\$3,044.04$/yr). (1) $P_{\text{phys}} = \$1.85 + \$21.50 = \$23.35$ (where \$20.00 = 1.20% p.a. mutual default pool on \$20k); (2) $\Delta_{\text{spread}} = \$253.67 - \$23.35 = \$230.32$; (3) $\gamma(0.25) = (1 - 0.25^2)^{-1/2} = 1.0328$; (4) $W_{\text{num}} = 1.0328 \times 1.04 \times 6.80 = 7.3039$; (5) $M_{\text{pure}} = 7.3039/0.96 = \$7.6083$; (6) $M_{\text{fair}} = \$7.6083/0.98 = \$7.7635$; (7) RRP/D = $\$23.35 + \$7.7635 = \$\mathbf{31.11}$/mo ($\$373.32$/yr); (8) $R_a = 1 - (7.6083/230.32) = \mathbf{96.7}\%$; (9) $\Delta P = \$253.67 - \$31.11 = \$\mathbf{222.56}$/mo ($\$2,670.72$/yr), $\Phi_{\text{fairness}} = \mathbf{12.3}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 40.99\%, \text{INDEX}_{\text{RRP/D,DSP}} = 5.03\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 35.96\%, \Delta\tau_{\text{NMW}} = \mathbf{505.04}$ min; (11) Effective Rate = $(\$3,044.04/\$20,000) \times 100\% = \mathbf{15.22}\%$ p.a.; Fair RRP/D Rate = $(\$373.32/\$20,000) \times 100\% = \mathbf{1.87}\%$ p.a.; RBA Spread = $+\mathbf{990}$ bps; Per \$10k Borrowed ($\times$0.5): Obs = $\$\mathbf{1,522.00}$/yr, Fair = $\$\mathbf{186.66}$/yr, Extractive Toll = $\$\mathbf{1,335.34}$/yr.

- **Scale 3 — Owner-Occupier Residential Mortgage (\$100,000 Tranche @ 6.46% p.a. + \$6.59/mo Package Fee):** Monthly Interest = $(\$100,000 \times 0.0646)/12 = \$538.33$; $P_{\text{retail}} = \$538.33 + \$6.59 = \$544.92$/mo ($\$6,539.04$/yr). (1) $P_{\text{phys}} = \$2.35 + \$12.10 = \$14.45$ (where \$10.00 = 0.12% p.a. collateralized loss pool per \$100k); (2) $\Delta_{\text{spread}} = \$544.92 - \$14.45 = \$530.47$; (3) $\gamma(0.22) = (1 - 0.22^2)^{-1/2} = 1.0251$; (4) $W_{\text{num}} = 1.0251 \times 1.05 \times 9.20 = 9.9026$; (5) $M_{\text{pure}} = 9.9026/0.98 = \$10.1047$; (6) $M_{\text{fair}} = \$10.1047/0.98 = \$10.3109$; (7) RRP/D = $\$14.45 + \$10.3109 = \$\mathbf{24.76}$/mo ($\$297.12$/yr); (8) $R_a = 1 - (10.1047/530.47) = \mathbf{98.1}\%$; (9) $\Delta P = \$544.92 - \$24.76 = \$\mathbf{520.16}$/mo ($\$6,241.92$/yr), $\Phi_{\text{fairness}} = \mathbf{4.5}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 88.05\%, \text{INDEX}_{\text{RRP/D,DSP}} = 4.00\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 84.05\%, \Delta\tau_{\text{NMW}} = \mathbf{1,180.39}$ min; (11) Effective Rate = $\mathbf{6.54}\%$ p.a.; Fair RRP/D Rate = $\mathbf{0.30}\%$ p.a.; RBA Spread = $+\mathbf{186}$ bps; Per \$10k Borrowed ($\times$0.1): Obs = $\$\mathbf{653.90}$/yr, Fair = $\$\mathbf{29.71}$/yr, Extractive Toll = $\$\mathbf{624.19}$/yr. *Standard \$500,000 Dwelling Mortgage ($\times$5 tranches):* Observed = $\$544.92 \times 5 = \$2,724.60$/mo ($\$32,695.20$/yr = $\$628.75$/wk); Fair RRP/D = $\$24.76 \times 5 = \$123.80$/mo ($\$1,485.60$/yr = $\$28.57$/wk); Extractive Toll = $\$520.16 \times 5 = \$\mathbf{2,600.80}$/mo ($\$\mathbf{31,209.60}$/yr = $\$\mathbf{600.18}$/wk = 96.98% of weekly DSP).

- **Scale 4 — Small Business (SME) & Agribusiness Secured Loan (\$100,000 Facility @ 7.72% p.a. + \$30.00/mo Line Fee):** Monthly Interest = $(\$100,000 \times 0.0772)/12 = \$643.33$; $P_{\text{retail}} = \$643.33 + \$30.00 = \$673.33$/mo ($\$8,079.96$/yr). (1) $P_{\text{phys}} = \$2.90 + \$32.10 = \$35.00$ (where \$29.17 = 0.35% p.a. secured SME default pool per \$100k) (2) $\Delta_{\text{spread}} = \$673.33 - \$35.00 = \$638.33$; (3) $\gamma(0.16) = 1.0131$; (4) $W_{\text{num}} = 1.0131 \times 1.06 \times 13.80 = 14.8189$; (5) $M_{\text{pure}} = 14.8189/0.95 = \$15.5989$; (6) $M_{\text{fair}} = \$15.5989/0.98 = \$15.9172$; (7) RRP/D = $\$35.00 + \$15.9172 = \$\mathbf{50.92}$/mo ($\$611.04$/yr); (8) $R_a = 1 - (15.5989/638.33) = \mathbf{97.6}\%$; (9) $\Delta P = \$673.33 - \$50.92 = \$\mathbf{622.41}$/mo ($\$7,468.92$/yr), $\Phi_{\text{fairness}} = \mathbf{7.6}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 108.80\%, \text{INDEX}_{\text{RRP/D,DSP}} = 8.23\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 100.58\%, \Delta\tau_{\text{NMW}} = \mathbf{1,412.43}$ min; (11) Effective Rate = $\mathbf{8.08}\%$ p.a.; Fair RRP/D Rate = $\mathbf{0.61}\%$ p.a.; RBA Spread

$= +\mathbf{312}$ bps; Per \$10k Borrowed ($\times 0.1$): Obs = $\mathbf{\$808.00}$/yr, Fair = $\mathbf{\$61.10}$/yr, Extractive Toll = $\mathbf{\$746.90}$/yr.

- **Scale 5 — Large Corporate Institutional Syndicate Loan (\$100,000 Tranche @ 5.79% p.a. + \$3.33/mo Agency Fee):** Monthly Interest = $(\$100,000 \times 0.0579)/12 = \$482.50$; $P_{\text{retail}} = \$482.50 + \$3.33 = \$485.83$/mo ($\$5,829.96$/yr). (1) $P_{\text{phys}} = \$1.60 + \$8.00 = \$9.60$ (where $\$6.67 = 0.08\%$ p.a. institutional default pool per \$100k); (2) $\Delta_{\text{spread}} = \$485.83 - \$9.60 = \$476.23$; (3) $\gamma(0.04) = 1.0008$; (4) $W_{\text{num}} = 1.0008 \times 1.02 \times 6.20 = 6.3291$; (5) $M_{\text{pure}} = 6.3291/0.98 = \$6.4582$; (6) $M_{\text{fair}} = \$6.4582/0.98 = \$6.5900$; (7) RRP/D = $\$9.60 + \$6.5900 = \mathbf{\$16.19}$/mo ($\$194.28$/yr); (8) $R_a = 1 - (6.4582/476.23) = \mathbf{98.6}\%$; (9) $\Delta P = \$485.83 - \$16.19 = \mathbf{\$469.64}$/mo ($\$5,635.68$/yr), $\Phi_{\text{fairness}} = \mathbf{3.3}\%$; (10) $\text{INDEX}_{\text{obs,DSP}} = 78.51\%, \text{INDEX}_{\text{RRP/D,DSP}} = 2.62\%, \Delta\text{INDEX}_{\text{toll,DSP}} = 75.89\%, \Delta\tau_{\text{NMW}} = \mathbf{1,065.75}$ min; (11) Effective Rate = $\mathbf{5.83}\%$ p.a.; Fair RRP/D Rate = $\mathbf{0.19}\%$ p.a.; RBA Spread = $+\mathbf{119}$ bps; Per \$10k Borrowed ($\times 0.1$): Obs = $\mathbf{\$583.00}$/yr, Fair = $\mathbf{\$19.43}$/yr, Extractive Toll = $\mathbf{\$563.57}$/yr. *Regressive Ratios:* Credit Card vs. Corporate Rate Multiplier = $21.71\%/5.83\% = \mathbf{3.72}\times$; Personal Loan vs. Corporate Rate Multiplier = $15.22\%/5.83\% = \mathbf{2.61}\times$; SME vs. Corporate Nominal Spread = $(7.72\% - 5.79\%) \times 100 = +\mathbf{193.0}$ bps.

8 Policy Implications & Regulatory Forensics

1. **Objective Pricing Benchmarks (RRP/D):** By replacing opaque corporate "cost-plus" claims with the UPE biophysical difficulty standard, regulatory authorities (such as the ACCC) can enforce price gouging caps anchored directly in physical difficulty.

2. **Sovereign Mutual Credit & Strategic Buffer Reserves:** Eliminating usurious working capital interest charges via Sovereign Mutual Credit Unions (SMCU) and establishing municipal buffer stocks collapses $R_a \to 0.02$, returning retail prices to their fair RRP/D baselines (Godley and Lavoie, 2007; Hamilton, 2026b).

3. **Decoupling from Speculative Indices:** The UPE proves that essential subsistence goods should never be priced via speculative financial exchanges (e.g., CME wheat futures, ICE sugar futures, MATIF canola oilseed, or global oil benchmarks), but rather via their local biophysical reproduction costs.

4. **Digital, Construction Supply Chain, Agrochemical & Healthcare Common Carrier Regulation:** The severe drag observed in NBN broadband (79.2%), clay bricks (62.8%), canola grain (62.6%), HDPE pipe (62.5%), gypsum plasterboard (60.3%), FIBC bulk bags (59.7%), urea fertilizer (58.8%), salbutamol inhalers (57.8%), solar salt (57.2%), glyphosate herbicide (51.3%), and steel roofing (50.4%) confirms that non-discretionary civic infrastructure, agricultural inputs, life-saving medical devices, and building envelope materials require statutory rate-of-return caps. Anchoring tariffs to physical RRP/D would immediately lower household broadband costs by 61%, rural water pipe costs by 32%, plasterboard linings by 32%, bulk urea fertilizer costs by 20%, and agricultural herbicide costs by 22%.

5. **Quantum States of Observer-Dependent Items & Services:** Beyond static scalar pricing, extending the Universal Price Equation into an observer-relative

quantum state formulation. An economic good or service (such as bread, fuel, or emergency care) is not an isolated classical scalar, but an entangled bipartite state vector whose experienced difficulty, confiscated labor time, and functional affordances dynamically transform relative to the observer's wealth coordinate (Y), budget exhaustion velocity (v_{rel}), and Lorentz coercion factor (γ).

9 Conclusion

This empirical case study demonstrates that the Universal Price Equation is not an abstract theoretical ideal, but an operational auditing instrument capable of diagnosing real-world supply chain sclerosis. By establishing that between 20% and 79% of Australian retail prices consist of unearned institutional rents, Value Physics provides a rigorous physical foundation for post-neoclassical economic reform.

CRediT Authorship Contribution Statement

Jarrod Lyndsay Hamilton: Conceptualization, Methodology, Formal Analysis, Software, Investigation, Data Curation, Writing – Original Draft, Visualization.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

All simulation scripts and calibration datasets are publicly available in the open-source repository (Hamilton, 2026b).

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Table 1: Master 70-Commodity Comparative Pricing, RRP/D & DSP INDEX Fairness Audit Matrix

Commodity	Sector	Unit	Obs. RRP	P_c	P_b	U (WEST)	Fair RRP/D	Rent ΔP	R_a (%)	Fairness	DSP INDEX
Sliced White Bread (680g)	Food	loaf	$4.10	$0.38	$0.45	2.15	**$3.03**	$1.07	34.2%	73.8%	0.66%
Milling Wheat (APW1)	Food	tonne	$365.00	$104.40	$82.00	95.00	**$290.28**	$74.72	43.0%	79.5%	58.98%
Full Cream Milk (1L)	Food	litre	$2.10	$0.31	$0.48	0.75	**$1.64**	$0.46	36.4%	78.1%	0.34%
Beef Cattle Weaners	Food	kg lwt	$3.85	$0.42	$1.15	1.25	**$3.32**	$0.53	25.0%	86.1%	0.62%
Diesel Fuel (ULSD)	Energy	litre	$2.15	$0.32	$0.92	0.35	**$1.67**	$0.48	53.5%	77.8%	0.35%
Grid Electricity Flat	Energy	kWh	$0.36	$0.08	$0.07	0.09	**$0.27**	$0.09	46.3%	73.7%	0.06%
LPG Cylinder (45kg)	Energy	45kg cyl	$165.00	$29.40	$58.36	28.50	**$126.26**	$38.74	51.1%	76.5%	26.66%
Bulk Water Cartage	Water	kL	$15.50	$4.32	$3.95	3.20	**$14.12**	$1.38	20.8%	91.1%	2.50%
Domestic Waste Lift	Sanitation	bin lift	$9.33	$2.45	$1.85	2.15	**$8.05**	$1.28	27.0%	86.3%	1.51%
Amoxicillin 500mg (20s)	Care	pack	$16.50	$2.10	$3.20	5.80	**$13.37**	$3.13	29.4%	81.0%	2.67%
Softwood Timber (MGP10)	Shelter	metre	$7.20	$0.85	$1.20	2.10	**$5.13**	$2.07	41.4%	71.2%	1.16%
Ready-Mix Concrete (N20)	Shelter	m^3	$295.00	$62.00	$78.00	58.00	**$221.72**	$73.28	48.3%	75.2%	47.67%
Steel Mesh (SL82)	Shelter	sheet	$128.00	$28.50	$36.80	22.50	**$97.18**	$30.82	50.2%	75.9%	20.68%
Bulk Road Freight	Logistics	nt-km	$0.1200	$0.0300	$0.0300	0.0200	**$0.0900**	$0.0400	59.1%	71.1%	0.02%
Merino Greasy Wool (19μm)	Agri-Fiber	kg clean	$15.50	$1.45	$2.20	5.60	**$10.82**	$4.68	40.7%	69.8%	2.50%
NBN Broadband (50/20)	Utilities	month	$85.00	$4.80	$14.50	12.20	**$33.22**	$51.78	79.2%	39.1%	13.74%
Salbutamol Inhaler (MDI)	Care	inhaler	$12.50	$0.65	$1.35	3.40	**$6.53**	$5.97	57.8%	52.2%	2.02%
Urban Reticulated Water	Water	kL	$3.60	$0.48	$0.62	1.15	**$2.82**	$0.78	32.5%	78.4%	0.58%
Broiler Chicken Whole	Food	kg	$4.50	$0.46	$1.68	1.15	**$3.45**	$1.05	45.8%	76.6%	0.73%
Clay Facing Bricks	Shelter	1k bricks	$1150.00	$278.00	$182.00	185.00	**$722.12**	$427.88	62.8%	62.8%	185.83%
Coarse Solar Sea Salt	Food/Min.	25kg bag	$14.50	$1.65	$2.10	4.20	**$8.45**	$6.05	57.2%	58.2%	2.34%
Raw Cane Sugar	Food	tonne	$620.00	$142.00	$168.00	145.00	**$487.77**	$132.23	43.8%	78.7%	100.19%
HDPE Water Pipe (100m)	Infra/Water	100m coil	$245.00	$48.50	$68.00	38.00	**$165.60**	$79.40	62.5%	67.6%	39.59%
Farmgate Canola Grain	Food/Oil	tonne	$685.00	$118.50	$215.00	112.00	**$467.77**	$217.23	62.6%	68.3%	110.69%
FIBC Bulk Bags (1-Tonne)	Logistics	bag	$24.50	$4.20	$5.90	4.80	**$16.02**	$8.48	59.7%	65.4%	3.96%
Bulk Portland Cement	Shelter	tonne	$345.00	$112.00	$84.00	56.00	**$278.99**	$66.01	45.4%	80.9%	55.75%
Corrugated Steel Roofing	Shelter	metre	$21.50	$4.85	$6.20	3.80	**$16.34**	$5.16	50.4%	76.0%	3.47%
Urea Fertilizer (46% N)	Agri/Chem	tonne	$740.00	$315.00	$165.00	78.00	**$589.25**	$150.75	58.8%	79.6%	119.58%
Glyphosate 450 SL (20L)	Agri/Chem	20L drum	$135.00	$22.50	$54.00	21.50	**$105.55**	$29.45	51.3%	78.2%	21.81%
Gypsum Plasterboard	Shelter	sheet	$24.80	$4.90	$6.40	4.10	**$16.77**	$8.03	60.3%	67.6%	4.01%
Hardwood Timber (F17)	Shelter	metre	$14.80	$2.15	$2.85	3.20	**$10.73**	$4.07	42.8%	72.5%	2.39%
Feed Barley (F1)	Food	tonne	$325.00	$92.50	$74.00	84.00	**$259.29**	$65.71	42.6%	79.8%	52.52%
Copper Building Wire	Energy/Infra	100m drum	$225.00	$24.50	$118.50	22.00	**$175.01**	$49.99	61.8%	77.8%	36.36%
Canola Protein Meal	Food/Feed	tonne	$580.00	$84.50	$212.00	82.00	**$394.91**	$185.09	66.0%	68.1%	93.72%
Universal Beam 150UB14	Shelter	metre	$52.00	$11.20	$14.50	8.20	**$37.28**	$14.72	56.9%	71.7%	8.40%
Whole Fuzzy Cottonseed	Food/Feed	tonne	$520.00	$86.50	$178.50	68.00	**$353.33**	$166.67	66.0%	67.9%	84.03%
Galvanized C-Purlin C15015	Shelter	metre	$16.50	$3.40	$5.10	2.60	**$12.11**	$4.39	55.8%	73.4%	2.67%
Prime Hard Wheat (APH2)	Food	tonne	$410.00	$118.50	$96.50	108.00	**$336.55**	$73.45	38.9%	82.1%	66.25%
Double-Glazed Window	Shelter	unit	$540.00	$52.00	$168.00	64.80	**$311.23**	$228.77	72.1%	57.6%	87.26%
Commercial Feed Sorghum	Food	tonne	$360.00	$104.50	$82.00	92.00	**$290.09**	$69.91	41.5%	80.6%	58.17%
Structural RHS Box 100x4	Shelter	metre	$42.20	$9.40	$12.60	6.80	**$31.60**	$10.60	53.4%	74.9%	6.82%
Desi Chickpeas (CHKP1)	Food/Pulse	tonne	$710.00	$135.00	$145.00	114.00	**$408.79**	$301.21	70.7%	57.6%	114.73%
Heavy Steel Plate 12mm	Shelter	m^2	$220.70	$52.50	$64.50	34.00	**$164.00**	$56.70	55.6%	74.3%	35.66%
Commercial Faba Beans	Food/Pulse	tonne	$455.00	$118.00	$98.00	98.00	**$318.99**	$136.01	57.8%	70.1%	73.52%
Equal Steel Angle 100x8	Shelter	metre	$43.50	$9.80	$12.90	6.90	**$32.27**	$11.23	54.9%	74.2%	7.03%
Australian Field Peas	Food/Pulse	tonne	$440.00	$112.50	$94.50	94.00	**$305.78**	$134.22	58.5%	69.5%	71.10%
Parallel Flange Channel	Shelter	metre	$65.50	$14.50	$19.80	10.20	**$48.44**	$17.06	55.6%	74.0%	10.58%
Narrow-Leaf Lupins	Food/Pulse	tonne	$520.00	$131.00	$121.00	112.00	**$368.55**	$151.45	57.4%	70.9%	84.03%
Built-in Steel Lintel 200x10	Shelter	metre	$105.00	$22.80	$33.60	16.50	**$79.36**	$25.64	53.7%	75.6%	16.97%
Australian Mungbeans	Food/Pulse	tonne	$1150.00	$175.00	$185.00	165.00	**$546.77**	$603.23	76.8%	47.5%	185.83%
Parallel Flange Channel 200x75	Shelter	metre	$85.00	$18.80	$25.60	13.20	**$62.70**	$22.30	55.8%	73.8%	13.74%
Split Faba Beans	Food/Pulse	tonne	$750.00	$226.00	$156.00	138.00	**$532.48**	$217.52	59.9%	71.0%	121.19%
Parallel Flange Channel 250x90	Shelter	metre	$132.00	$29.20	$39.70	20.40	**$97.19**	$34.81	56.1%	73.6%	21.33%
Split Yellow Field Peas	Food/Pulse	tonne	$680.00	$207.00	$145.00	132.00	**$493.28**	$186.72	57.8%	72.5%	109.88%
Parallel Flange Channel 300x90	Shelter	metre	$149.50	$33.20	$45.20	23.10	**$110.43**	$39.07	55.8%	73.9%	24.16%
Split Red Lentils	Food/Pulse	tonne	$1050.00	$262.00	$178.00	146.00	**$603.57**	$446.43	73.7%	57.5%	169.67%
Saline IV Infusion 1,000mL	Care	bag	$8.50	$0.65	$0.85	2.40	**$4.99**	$3.51	51.2%	58.7%	1.37%
Solar PV Module 440W	Energy/Shelter	panel	$165.00	$38.50	$52.00	14.50	**$108.79**	$56.21	75.9%	65.9%	26.66%
Universal Column 200UC46	Shelter	metre	$172.00	$38.50	$52.00	26.50	**$127.25**	$44.75	55.8%	74.0%	27.79%
Faba Protein Concentrate 55%	Food/Pulse	tonne	$1280.00	$310.00	$245.00	162.00	**$736.38**	$543.62	75.5%	57.5%	206.84%
Universal Column 250UC73	Shelter	metre	$272.00	$60.50	$82.50	41.50	**$200.55**	$71.45	56.3%	73.7%	43.95%
Field Pea Protein Isolate 80%	Food/Pulse	tonne	$3450.00	$820.00	$615.00	385.00	**$1866.05**	$1583.95	79.0%	54.1%	557.49%
Universal Column 310UC97	Shelter	metre	$360.00	$80.50	$109.50	55.00	**$266.27**	$93.73	56.0%	74.0%	58.17%
Chickpea Protein Isolate 75%	Food/Pulse	tonne	$3150.00	$745.00	$560.00	350.00	**$1700.63**	$1449.37	79.0%	54.0%	509.01%
Universal Column 310UC118	Shelter	metre	$438.00	$98.00	$133.50	67.00	**$324.41**	$113.59	55.9%	74.1%	70.78%
Faba Protein Isolate 85%	Food/Pulse	tonne	$3580.00	$855.00	$640.00	398.00	**$1940.60**	$1639.40	79.1%	54.2%	578.49%
Universal Column 310UC158	Shelter	metre	$588.00	$131.50	$179.50	90.00	**$435.80**	$152.20	55.8%	74.1%	95.01%
Lupin Protein Isolate 88%	Food/Pulse	tonne	$3750.00	$895.00	$670.00	415.00	**$2016.73**	$1733.27	79.7%	53.8%	605.96%
Universal Beam (200UB18.2)	Shelter	metre	$68.50	$15.20	$20.80	10.60	**$50.83**	$17.67	55.3%	74.2%	11.07%
Oat Milk Base (15% Beta-Glucan)	Food/Grain	tonne	$2450.00	$580.00	$440.00	285.00	**$1339.35**	$1110.65	78.1%	54.7%	395.90%

Table 2: 14-Item Weekly Household Subsistence Basket & DSP Subsistence Floor INDEX Audit (Y_{DSP} = \$618.85/wk)

Basket Necessity	Weekly Qty	Obs. RRP	Fair RRP/D	Drag Toll ΔP	Obs. DSP INDEX	Fair DSP INDEX	DSP Toll INDEX	NMW Stolen Time
Commercial Sliced White Bread	2.5 loaves/wk	$10.25	$7.57	$2.68	1.66%	1.22%	0.43%	6.9 min
Fresh Full Cream Milk	4.0 litres/wk	$8.40	$6.56	$1.84	1.36%	1.06%	0.30%	4.7 min
Commercial Broiler Chicken	2.0 kg/wk	$9.00	$6.90	$2.10	1.45%	1.11%	0.34%	5.4 min
Automotive Diesel Fuel	45.0 litres/wk	$96.75	$75.15	$21.60	15.63%	12.14%	3.49%	55.4 min
Residential Grid Electricity	115.0 kWh/wk	$41.40	$31.05	$10.35	6.69%	5.02%	1.67%	26.5 min
LPG Bottled Gas	0.2 cylinders/wk	$33.00	$25.25	$7.75	5.33%	4.08%	1.25%	19.9 min
Urban Reticulated Potable Water	4.5 kL/wk	$16.20	$12.69	$3.51	2.62%	2.05%	0.57%	9.0 min
Domestic Waste Collection	1.0 lift/wk	$9.33	$8.05	$1.28	1.51%	1.30%	0.21%	3.3 min
Domestic Broadband Data Transit	0.2308 monthly plan/wk	$19.62	$7.67	$11.95	3.17%	1.24%	1.93%	30.7 min
Prescription Antibiotic	0.1 course/wk	$1.65	$1.34	$0.31	0.27%	0.22%	0.05%	0.8 min
Prescription Salbutamol Inhaler	0.15 inhaler/wk	$1.88	$0.98	$0.90	0.30%	0.16%	0.15%	2.3 min
Residential Structural Softwood Timber	2.0 m/wk maint	$14.40	$10.26	$4.14	2.33%	1.66%	0.67%	10.6 min
Corrugated Galvanized Steel Roofing Sheets	0.5 m/wk maint	$10.75	$8.17	$2.58	1.74%	1.32%	0.42%	6.6 min
Standard Gypsum Plasterboard Wall Lining	0.5 sheet/wk maint	$12.40	$8.38	$4.02	2.00%	1.35%	0.65%	10.3 min
Total Weekly Basket (14 Items)	**Weekly Bundle**	**$285.02**	**$210.03**	**$74.99**	**46.06%**	**33.94%**	**12.12%**	**170.2 min (2.84 hr)**

Table 3: Big 4 Australian Bank Loan Fairness & Multi-Scale Credit Spread Matrix (RBA Cash Rate = 4.60% p.a.)

Credit Scale & Facility	Nom./Eff. Rate	Fair Rate	RBA Spread	Obs./Mo	Fair/Mo	Toll/Mo	Obs./$10k/Yr	Fair/$10k/Yr	R_a (%)	Fairness
1. Individual Credit Card ($5k)	20.99% / 21.71%	**3.20%**	+1,639 bps	$90.46	**$13.32**	$77.14	$2,171.00	$319.68	95.3%	14.7%
2. Unsecured Personal Loan ($20k)	14.50% / 15.22%	**1.87%**	+990 bps	$253.67	**$31.11**	$222.56	$1,522.00	$186.66	96.7%	12.3%
3. Owner-Occupier Mortgage ($100k)	6.46% / 6.54%	**0.30%**	+186 bps	$544.92	**$24.76**	$520.16	$653.90	$29.71	98.1%	4.5%
4. SME & Agribusiness Loan ($100k)	7.72% / 8.08%	**0.61%**	+312 bps	$673.33	**$50.92**	$622.41	$808.00	$61.10	97.6%	7.6%
5. Large Corporate Syndicate ($100k)	5.79% / 5.83%	**0.19%**	+119 bps	$485.83	**$16.19**	$469.64	$583.00	$19.43	98.6%	3.3%

Table 4: Complete 70-Commodity Intermediate Mathematical Working-Out & Derivation Ledger ($P_{\text{phys}}, \Delta_{\text{spread}}, \gamma, W_{\text{num}}, M_{\text{pure}}, M_{\text{fair}}, \text{RRP/D}, R_a$)

Commodity	P_{retail}	$P_{\text{phys}} = P_c + P_b$	Δ_{spread}	$\gamma(v_{\text{rel}})$	$W_{\text{num}} = \gamma SU$	$M_{\text{pure}} = W_{\text{num}}/R_n$	$M_{\text{fair}} = M_{\text{pure}}/0.98$	Fair RRP/D	Drag R_a	Premium ΔP	Fairness
1. Commercial Sliced White Bread (680g)	$4.10	$0.83	$3.27	1.0013	2.15	$2.15	$2.20	**$3.03**	34.2%	$1.07	73.8%
2. Milling Wheat (APW1 / ASW1)	$365.00	$186.40	$178.60	1.0002	99.77	$101.81	$103.88	**$290.28**	43.0%	$74.72	79.5%
3. Fresh Full Cream Milk (Full Fat)	$2.10	$0.79	$1.31	1.0050	0.79	$0.83	$0.85	**$1.64**	36.4%	$0.46	78.1%
4. Farmgate Beef Cattle Weaners (British Breed)	$3.85	$1.57	$2.28	1.0114	1.45	$1.71	$1.75	**$3.32**	25.0%	$0.53	86.1%
5. Automotive Diesel Fuel (Ultra-Low Sulphur)	$2.15	$1.24	$0.91	1.0073	0.38	$0.42	$0.43	**$1.67**	53.5%	$0.48	77.8%
6. Residential Grid Electricity (Single-Rate Flat)	$0.36	$0.15	$0.21	1.0032	0.10	$0.11	$0.12	**$0.27**	46.3%	$0.09	73.7%
7. LPG Bottled Gas (45kg Domestic Cylinder)	$165.00	$87.76	$77.24	1.0131	33.20	$37.73	$38.50	**$126.26**	51.1%	$38.74	76.5%
8. Bulk Potable Water Cartage (Regional NSW)	$15.50	$8.27	$7.23	1.0328	4.30	$5.73	$5.85	**$14.12**	20.8%	$1.38	91.1%
9. Domestic Waste Collection (Regional NSW)	$9.33	$4.30	$5.03	1.0251	2.64	$3.67	$3.75	**$8.05**	27.0%	$1.28	86.3%
10. Prescription Antibiotic (Amoxicillin 500mg, 20 Caps)	$16.50	$5.30	$11.20	1.0675	7.43	$7.90	$8.07	**$13.37**	29.4%	$3.13	81.0%
11. Residential Structural Softwood Timber (MGP10 90x45mm Pine)	$7.20	$2.05	$5.15	1.0114	2.65	$3.02	$3.08	**$5.13**	41.4%	$2.07	71.2%
12. Pre-Mixed Ready-Mix Concrete (N20 20 MPa Structural)	$295.00	$140.00	$155.00	1.0206	68.08	$80.09	$81.72	**$221.72**	48.3%	$73.28	75.2%
13. Structural Steel Reinforcing Mesh (SL82, 6.0m x 2.4m Sheet)	$128.00	$65.30	$62.70	1.0166	25.62	$31.24	$31.88	**$97.18**	50.2%	$30.82	75.9%
14. Bulk Road Freight Transport (Newell Highway B-Double Corridor)	$0.12	$0.06	$0.06	1.0099	0.02	$0.03	$0.03	**$0.09**	59.1%	$0.04	71.1%
15. Australian Merino Greasy Wool (19-Micron Fleece)	$15.50	$3.65	$11.85	1.0032	6.18	$7.02	$7.17	**$10.82**	40.7%	$4.68	69.8%
16. Domestic Broadband Data Transit (NBN 50/20 Mbps Tier)	$85.00	$19.30	$65.70	1.0114	12.96	$13.64	$13.92	**$33.22**	79.2%	$51.78	39.1%
17. Prescription Salbutamol Inhaler (100mcg CFC-Free MDI, 200 Doses)	$12.50	$2.00	$10.50	1.0911	4.08	$4.44	$4.53	**$6.53**	57.8%	$5.97	52.2%
18. Urban Reticulated Potable Water (Municipal Pipeline Tariff)	$3.60	$1.10	$2.50	1.0206	1.35	$1.69	$1.72	**$2.82**	32.5%	$0.78	78.4%
19. Commercial Broiler Chicken (Whole Fresh Dressed Bird)	$4.50	$2.14	$2.36	1.0073	1.22	$1.28	$1.31	**$3.45**	45.8%	$1.05	76.6%
20. Standard Extruded Clay Facing Bricks (Common Structural Masonry)	$1150.00	$460.00	$690.00	1.0166	210.64	$256.88	$262.12	**$722.12**	62.8%	$427.88	62.8%
21. Coarse Solar Sea Salt (Unrefined Food & Livestock Grade)	$14.50	$3.75	$10.75	1.0018	4.42	$4.60	$4.70	**$8.45**	57.2%	$6.05	58.2%
22. Raw Centrifugal Cane Sugar (Bulk Crystal Milling Feedstock)	$620.00	$310.00	$310.00	1.0013	156.80	$174.22	$177.77	**$487.77**	43.8%	$132.23	78.7%
23. HDPE Rural Water Reticulation Pipe (32mm PN12.5, 100m Coil)	$245.00	$116.50	$128.50	1.0131	42.35	$48.12	$49.10	**$165.60**	62.5%	$79.40	67.6%
24. Farmgate Canola Grain (Bulk Non-GM Oilseed)	$685.00	$333.50	$351.50	1.0008	121.06	$131.58	$134.27	**$467.77**	62.6%	$217.23	68.3%
25. Commercial FIBC Polypropylene Bulk Bags (1-Tonne Woven Container)	$24.50	$10.10	$14.40	1.0073	5.22	$5.80	$5.92	**$16.02**	59.7%	$8.48	65.4%
26. Bulk Portland Cement (General Purpose GP Clinker)	$345.00	$196.00	$149.00	1.0114	63.44	$81.33	$82.99	**$278.99**	45.4%	$66.01	80.9%
27. Corrugated Galvanized Steel Roofing Sheets (Custom Orb 0.42mm BMT)	$21.50	$11.05	$10.45	1.0166	4.25	$5.18	$5.29	**$16.34**	50.4%	$5.16	76.0%
28. Bulk Granular Agricultural Urea Fertilizer (46% Nitrogen)	$740.00	$480.00	$260.00	1.0114	89.94	$107.07	$109.25	**$589.25**	58.8%	$150.75	79.6%
29. Agricultural Glyphosate Herbicide (450g/L SL Systemic Formulation)	$135.00	$76.50	$58.50	1.0166	24.48	$28.46	$29.05	**$105.55**	51.3%	$29.45	78.2%
30. Standard Gypsum Plasterboard Wall Lining (10mm Recessed Edge)	$24.80	$11.30	$13.50	1.0099	4.55	$5.36	$5.47	**$16.77**	60.3%	$8.03	67.6%
31. Commercial Structural Hardwood Timber (F17 Kiln-Dried Hardwood, 90x45mm)	$14.80	$5.00	$9.80	1.0099	4.04	$5.61	$5.73	**$10.73**	42.8%	$4.07	72.5%
32. Agricultural Feed Barley (F1 Broadacre Feed Grain)	$325.00	$166.50	$158.50	1.0008	89.11	$90.93	$92.79	**$259.29**	42.6%	$65.71	79.8%
33. Commercial Copper Electrical Building Wire (2.5mm2 Twin & Earth Flat TPS Cable)	$225.00	$143.00	$82.00	1.0166	25.72	$31.37	$32.01	**$175.01**	61.8%	$49.99	77.8%
34. Solvent-Extracted Canola Meal / Protein Cake (Bulk Pelletized, 36%-38% Crude Protein)	$580.00	$296.50	$283.50	1.0050	90.65	$96.44	$98.41	**$394.91**	66.0%	$185.09	68.1%
35. Commercial Hot-Rolled Structural Steel Universal Beams (150UB14.0)	$52.00	$25.70	$26.30	1.0131	9.30	$11.35	$11.58	**$37.28**	56.9%	$14.72	71.7%
36. Whole Ginned Cottonseed (Fuzzy White Cattle Feedstock)	$520.00	$265.00	$255.00	1.0050	77.91	$86.57	$88.33	**$353.33**	66.0%	$166.67	67.9%
37. Commercial Cold-Formed Steel Galvanized Framing / C-Purlin (C15015, 1.5mm BMT)	$16.50	$8.50	$8.00	1.0131	2.90	$3.53	$3.61	**$12.11**	55.8%	$4.39	73.4%
38. Australian Prime Hard Breadmaking Wheat (APH2 High-Protein Milling Grain, min 13% Protein)	$410.00	$215.00	$195.00	1.0008	116.73	$119.12	$121.55	**$336.55**	38.9%	$73.45	82.1%
39. Commercial Aluminum Window Glazing Frames (Double-Glazed Residential Sliding Window, 1200mm x 1800mm Standard Unit)	$540.00	$220.00	$320.00	1.0166	75.10	$89.40	$91.23	**$311.23**	72.1%	$228.77	57.6%
40. Commercial Feed Sorghum (SGM1 Summer Grain Sorghum, min 9% Protein)	$360.00	$186.50	$173.50	1.0013	99.48	$101.51	$103.59	**$290.09**	41.5%	$69.91	80.6%
41. Commercial Structural Steel Hollow Sections (100x100x4.0mm RHS/SHS, Grade C350LO)	$42.20	$22.00	$20.20	1.0131	7.72	$9.41	$9.60	**$31.60**	53.4%	$10.60	74.9%
42. Australian Desi Chickpeas (CHKP1 Premium Bulk Pulse Grain)	$710.00	$280.00	$430.00	1.0013	131.26	$126.22	$128.79	**$408.79**	70.7%	$301.21	57.6%
43. Commercial Heavy Structural Steel Plate (Grade 250, 12mm Medium Plate, AS/NZS 3678)	$220.70	$117.00	$103.70	1.0099	37.77	$46.06	$47.00	**$164.00**	55.6%	$56.70	74.3%
44. Commercial Faba Beans (F1 Broadacre Pulse Grain, min 24% Protein)	$455.00	$216.00	$239.00	1.0013	105.97	$100.93	$102.99	**$318.99**	57.8%	$136.01	70.1%
45. Commercial Structural Steel Equal Angle (100x100x8.0mm EA, Grade 300PLUS / AS/NZS 3679.1)	$43.50	$22.70	$20.80	1.0131	7.69	$9.38	$9.57	**$32.27**	54.9%	$11.23	74.2%
46. Australian Field Peas (Kapa Type Bulk Grain / Livestock Protein Feed, min 22% Protein)	$440.00	$207.00	$233.00	1.0013	101.65	$96.81	$98.78	**$305.78**	58.5%	$134.22	69.5%
47. Commercial Structural Steel Parallel Flange Channel (150x75mm PFC, Grade 300PLUS / AS/NZS 3679.1)	$65.50	$34.30	$31.20	1.0131	11.37	$13.86	$14.14	**$48.44**	55.6%	$17.06	74.0%
48. Australian Narrow-Leaf Lupins (Angustifolius Bulk Feed Lupins, min 32% Protein)	$520.00	$252.00	$268.00	1.0013	123.35	$114.22	$116.55	**$368.55**	57.4%	$151.45	70.9%
49. Commercial Galvanized Steel Built-In Lintel / T-Bar (200x10mm Lintel, AS/NZS 2699.3 / AS 4100)	$105.00	$56.40	$48.60	1.0166	18.45	$22.50	$22.96	**$79.36**	53.7%	$25.64	75.6%
50. Australian Mungbeans (Berken / Crystal Type Pulse Grain, min 24% Protein)	$1150.00	$360.00	$790.00	1.0032	190.36	$183.04	$186.77	**$546.77**	76.8%	$603.23	47.5%
51. Commercial Structural Steel Parallel Flange Channel (200x75mm PFC, Grade 300PLUS / AS/NZS 3679.1)	$85.00	$44.40	$40.60	1.0131	14.71	$17.94	$18.30	**$62.70**	55.8%	$22.30	73.8%
52. Commercial De-Hulled Split Faba Beans (Machine Dressed Export & Food Grade, min 28% Protein)	$750.00	$382.00	$368.00	1.0018	154.84	$147.47	$150.48	**$532.48**	59.9%	$217.52	71.0%
53. Commercial Structural Steel Parallel Flange Channel (250x90mm PFC, Grade 300PLUS / AS/NZS 3679.1)	$132.00	$68.90	$63.10	1.0131	22.73	$27.72	$28.29	**$97.19**	56.1%	$34.81	73.6%
54. Commercial De-Hulled Split Yellow Field Peas (Machine Dressed Food Grade, min 25% Protein)	$680.00	$352.00	$328.00	1.0013	145.38	$138.46	$141.28	**$493.28**	57.8%	$186.72	72.5%
55. Commercial Structural Steel Parallel Flange Channel (300x90mm PFC, Grade 300PLUS / AS/NZS 3679.1)	$149.50	$78.40	$71.10	1.0131	25.74	$31.39	$32.03	**$110.43**	55.9%	$39.07	73.9%
56. Commercial De-Hulled Split Red Lentils (Machine Dressed Export Grade, min 26% Protein)	$1050.00	$440.00	$610.00	1.0025	168.31	$160.30	$163.57	**$603.57**	73.7%	$446.43	57.5%
57. Pharmaceutical Saline IV Infusion Bag (Sodium Chloride 0.9%, 1,000 mL)	$8.50	$1.50	$7.00	1.0675	3.07	$3.42	$3.49	**$4.99**	51.2%	$3.51	58.7%
58. Commercial Solar PV Module (440W Monocrystalline Residential Panel, Tier-1)	$165.00	$90.50	$74.50	1.0073	15.77	$17.93	$18.29	**$108.79**	75.9%	$56.21	65.9%
59. Commercial Structural Steel Universal Column (200UC46.2 Grade 300PLUS)	$172.00	$90.50	$81.50	1.0131	29.53	$36.01	$36.75	**$127.25**	55.8%	$44.75	74.0%
60. Commercial Australian Faba Bean Protein Concentrate (min 55% Protein)	$1280.00	$555.00	$725.00	1.0018	186.64	$177.75	$181.38	**$736.38**	75.5%	$543.62	57.5%
61. Commercial Structural Steel Universal Column (250UC72.9 Grade 300PLUS)	$272.00	$143.00	$129.00	1.0131	46.25	$56.40	$57.55	**$200.55**	56.3%	$71.45	73.7%
62. Commercial Australian Field Pea Protein Isolate (min 80% Protein)	$3450.00	$1435.00	$2015.00	1.0018	443.55	$422.43	$431.05	**$1866.05**	79.0%	$1583.95	54.1%
63. Commercial Structural Steel Universal Column (310UC96.8 Grade 300PLUS)	$360.00	$190.00	$170.00	1.0131	61.29	$74.74	$76.27	**$266.27**	56.0%	$93.73	74.0%
64. Commercial Australian Chickpea Protein Isolate (min 75% Protein)	$3150.00	$1305.00	$1845.00	1.0018	403.23	$387.72	$395.63	**$1700.63**	79.0%	$1449.37	54.0%
65. Commercial Structural Steel Universal Column (310UC118 Grade 300PLUS)	$438.00	$231.50	$206.50	1.0131	74.66	$91.05	$92.91	**$324.41**	55.9%	$113.59	74.1%
66. Commercial Australian Faba Bean Protein Isolate (min 85% Protein)	$3580.00	$1495.00	$2085.00	1.0018	458.53	$436.69	$445.60	**$1940.60**	79.1%	$1639.40	54.2%
67. Commercial Structural Steel Universal Column (310UC158 Grade 300PLUS)	$588.00	$311.00	$277.00	1.0131	100.29	$122.31	$124.80	**$435.80**	55.8%	$152.20	74.1%
68. Commercial Australian Lupin Protein Isolate (min 88% Protein)	$3750.00	$1565.00	$2185.00	1.0018	478.11	$442.70	$451.73	**$2016.73**	79.7%	$1733.27	53.8%
69. Commercial Structural Steel Universal Beam (200UB18.2 Grade 300PLUS)	$68.50	$36.00	$32.50	1.0131	11.92	$14.54	$14.83	**$50.83**	55.3%	$17.67	74.2%
70. Commercial Australian Oat Milk Base (15% Beta-Glucan)	$2450.00	$1020.00	$1430.00	1.0018	325.49	$312.97	$319.35	**$1339.35**	78.1%	$1110.65	54.7%